

CONCEPT OF AROMATICITY

- evolved as the understanding of reasons for the properties of benzene and other aromatic molecules deepened.
- originally associated with a special reactivity.
- aromatic hydrocarbons were considered as those unsaturated systems that underwent substitution reactions in preference to addition.
- idea of special stability plays a big role.
- aromaticity is now generally associated with the property of lowered molecular energy because benzene can be shown to be much lower in ~~enthalpy~~ ^{energy} than predicted by summation of the normal bond energies for the $C=C$, $C-C$ and $C-H$ bonds in the Kekulé's representation of benzene.
- A major contribution to the

Stability of The aromatic systems is considered to be the delocalization of electrons in these molecules.

- Aromaticity is generally described in terms of molecular orbital (MO) terminology.

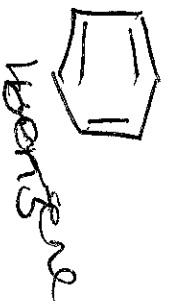
i.e. cyclic structures with particular stable arrangement of occupied π molecular orbitals are called aromatic.

- a sample expression of the relationship b/w aromatic orbital description of structure and aromaticity is known as the Hückel rule.

- derived from Hückel molecular orbital theory that states that:- a planar monocyclic, completely conjugated hydrocarbon will be aromatic when the ring contains $(4n+2)\pi$ electrons.

AROMATIC COMPOUNDS

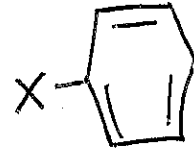
- are benzene and compounds that resemble benzene in nature.
- π - π^* are compounds that seem to differ from benzene in structure ~~but~~ yet which behave very much like benzene.
- π - π^* are expected to meet the $4n+2$ Hückel rule.
- π - π^* are flat or nearly flat molecules.
- Aromaticity can be determined by ~~atom~~ NMR spectrum.
 - if protons are shifted down - field from ~~the~~ normal olefinic region we can say that the molecule is aromatic.
- aromatic compounds have a sextate ~~atom~~ of electrons.



- Benzene along with some other compounds
eg heterocycles are aromatic.

MONOSUBSTITUTED AROMATIC COMPOUNDS

General formulae



PREPARATION

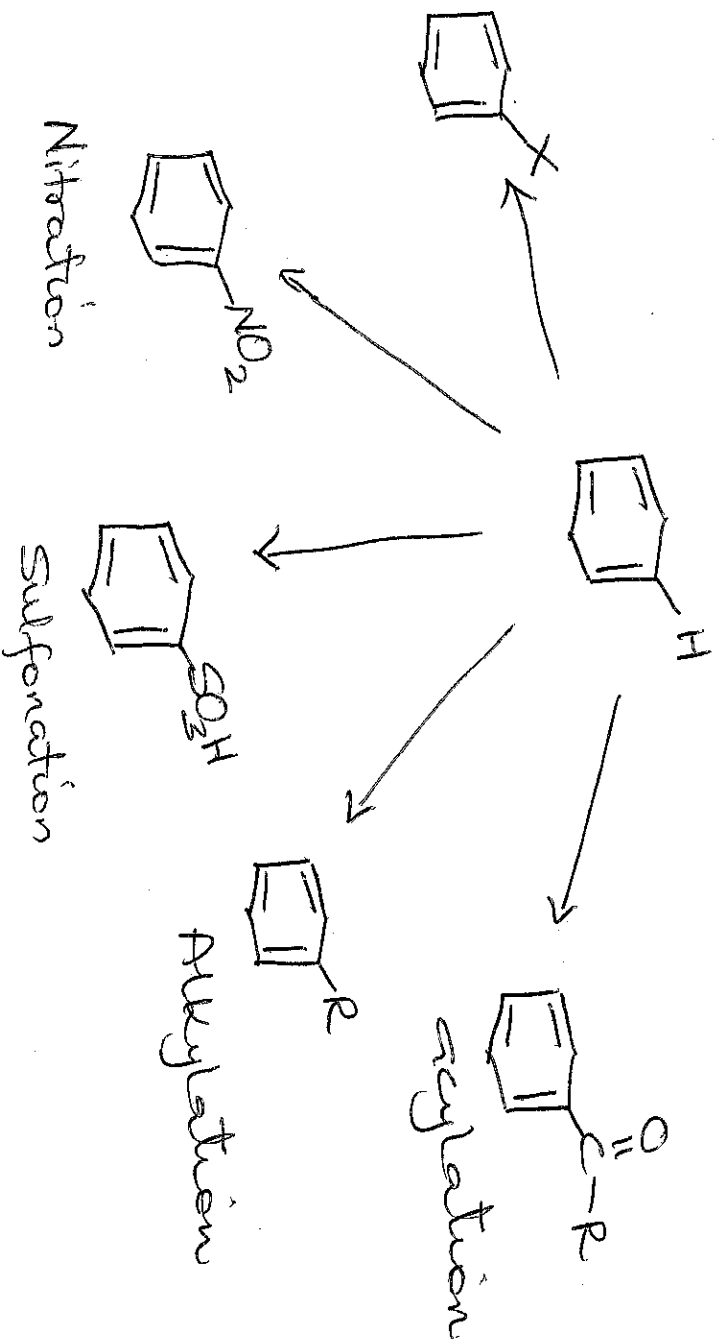
monosubstituted aromatic compounds are prepared by electrophilic aromatic substitution reactions.

- Electrophilic aromatic substitution is the most important rxn of aromatic compounds.

an electrophile (E^+) reacts with an aromatic ring and substitutes one of the hydrogens.

(2b)

—many diff substituents can be introduced on the aromatic ring by electrophilic substitution rxn. By choosing proper reagents, it's possible to halogenate (substitute a halogen: $-F, -Cl, -Br, \text{ or } -I$), nitrate (substitute a nitro group: $-\text{NO}_2$), sulphonate (substitute a sulphonic acid group: $-\text{SO}_3\text{H}$), alkylate (substitute and alkyl group: $-\text{R}$), or acylate (substitute an acyl group: $-\text{COR}$).



1. BROMINATION

Benzene reacts with bromine in the Br_2 in the presence of FeBr_3 as catalyst to yield bromobenzene



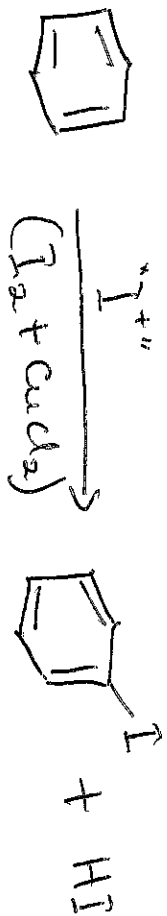
2. CHLORINATION



3. IODINATION

~~I₂~~ Iodine by itself is unreactive towards aromatic rings and a promoter is required for aromatic iodination. The best promoters are oxidizing agents such as hydrogen peroxide, H_2O_2 , or copper salts such as CuCl_2 . These promoters work by oxidizing I_2 to a more powerful electrophilic species that reacts as if it were I^+ . The benzene ring then attacks I^+ in the typical way.

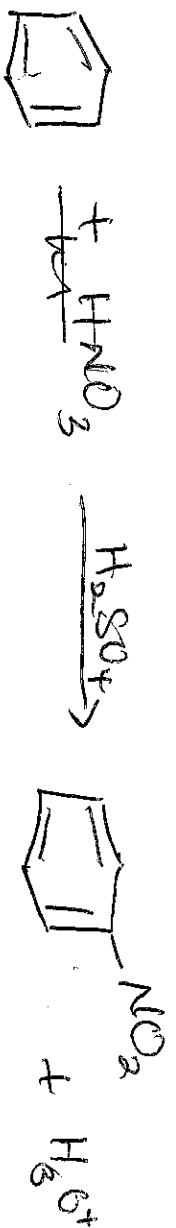
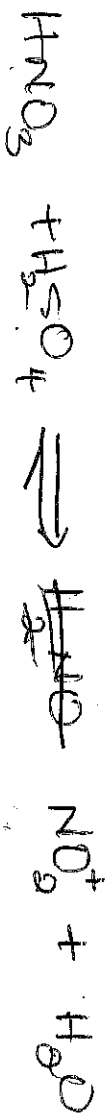
(3)



Fluorine is too reactive, and poor yields of monofluoroaromatic products are obtained.

4. NITRATION

Aromatic ring can be nitrated with a mixture of conc HNO_3 and H_2SO_4 acids. The electrophile in this reaction is the nitronium ion, NO_2^+ , which is generated from HNO_3 by protonation and loss of water.



Nitro benzenes
~~they~~ - can be reduced by reagents such as iron metal or SnCl_2 to yield aniline ArNH_2 .

-nitro aromatic compounds is a key part of the industrial synthesis of dyes and many pharmaceutical agents.

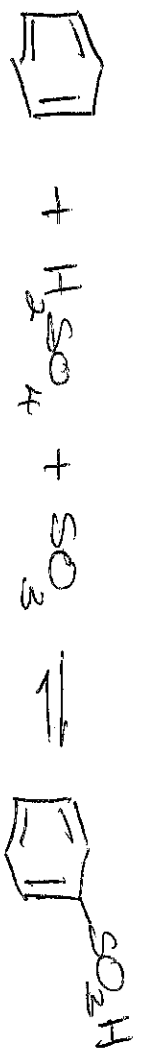
5. SULPHONATION

Aromatic rings can be sulphonated by reaction with fuming sulphuric acid, a mixture of H_2SO_4 and SO_3 . The reactive electrophile is either HSO_3^+ or neutral SO_3 , depending on reaction conditions.

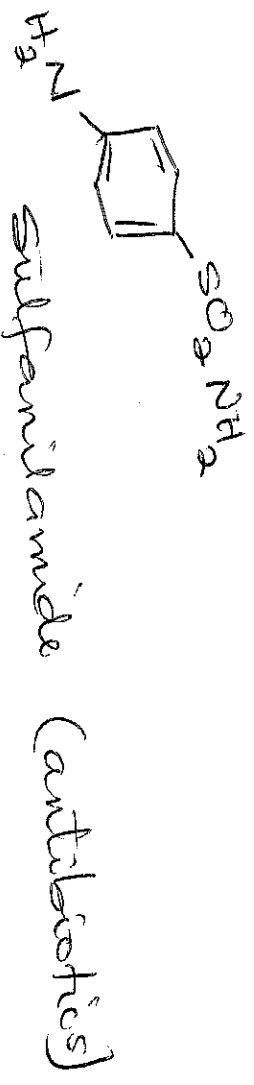
Sulphonation reaction is readily reversible, it can occur either forward or backward, depending on the reaction conditions. Sulphonation is favoured in strong acid, but desulphonation is favoured in hot, dil aqueous acid.



(4)



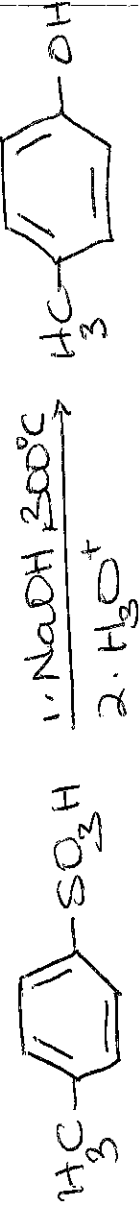
Aromatic Sulphonic acids are valuable intermediates in the preparation of dyes and pharmaceuticals. eg Sulfa drugs such as sulfanilamide, were among the first useful antibiotics known. Although largely replaced today by more effective agents, Sulfa Drugs are still used in the treatment of meningitis and urinary-tract infections. These drugs are prepared commercially by a process that involves aromatic sulfonation as the key step.



~~Sulfa~~ Aromatic Sulphonic acids are useful because of the further

chemistry they undergo.

Heating an aromatic sulfonic acid with NaOH at 300°C in the absence of solvent effects a replacement of the $-\text{SO}_3\text{H}$ group by $-\text{OH}$ and gives phenol. Yields of this so called alkali fusion reaction are generally good but the conditions are so vigorous that the rxn is not compatible with the presence of substituents other than alkyl groups on the aromatic ring.

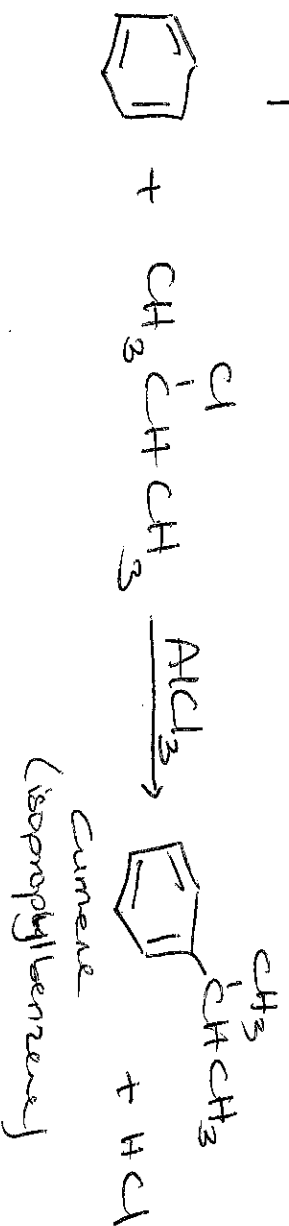


p-Cresol
(a phenol)

(5)

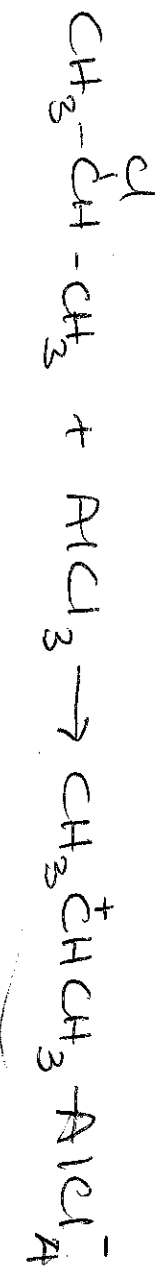
b. ALKYLATION (THE FRIEDEL-CRAFTS REACTION)

one of the most useful of all the electrophilic aromatic substitution reactions is ~~the~~ ^{alkylation}, the attachment of an alkyl group to the benzene ring.



— in alkylation, the electrophile is a carbocation, R^+ . AlCl_3 is a catalyst that helps the alkyl halide to ionize ~~so~~ ^{much} ~~the~~ ^{the} same way ~~as~~ that

FeBr_3 catalyzes aromatic bromination by polarizing the Br_2 .



Though Alkylation rxns is broadly used in the synthesis of alkyl benzenes. Friedel crafts alkylations have strict limitations.

① Only alkyl halides can be used. Alkyl fluorides, chlorides, bromides and iodides all react.

② Aryl halides and vinylic halides do not react

③ Aryl and vinylic ~~car~~ carbocations are too high in energy to form under Friedel Crafts conditions.



an aryl halide a vinylic halide



not reactive

(6)

(2) Friedel-Crafts reactions don't succeed on aromatic rings that are substituted by strongly electron-withdrawing groups.

Note: The presence of a substituent group already on a ring can have a dramatic effect on that ring's subsequent reactivity towards further electrophilic substitution. Rings that contain the following substituents are not reactive enough to undergo Friedel-Crafts alkylation.



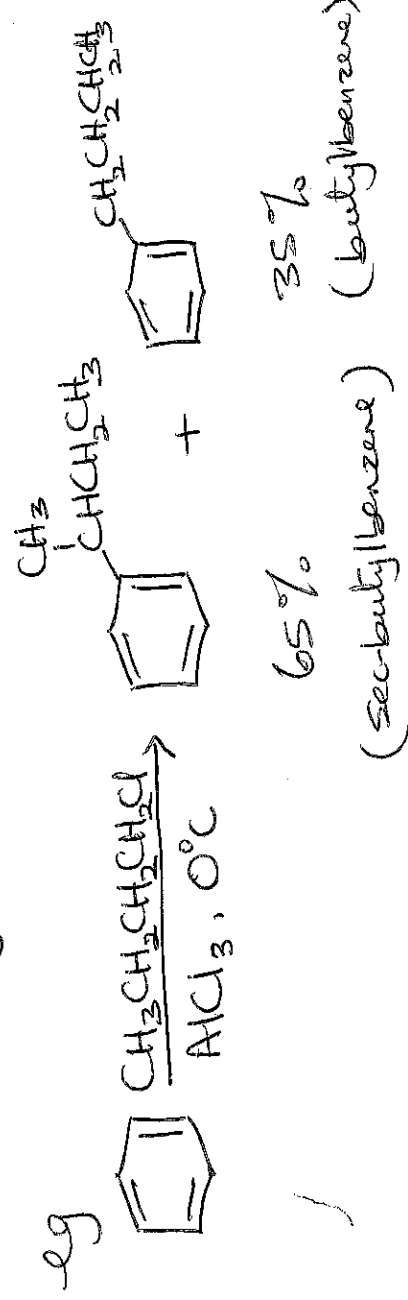
where $Y = -NR_3^+$, $-NO_2$, $-CN$, $-SO_3H$, $-CHO$, $-COCH_3$, $-COOH$, $-COOCH_3$, $(-NH_2, -NHR, -NR_2)$.

(3) It is often difficult to stop a reaction after a single substitution. Once an alkyl group is on the ring a second substitution is facilitated. Polyalkylation is often observed.

Large amounts of monoalkylation products are obtained only when a large excess of benzene is used.

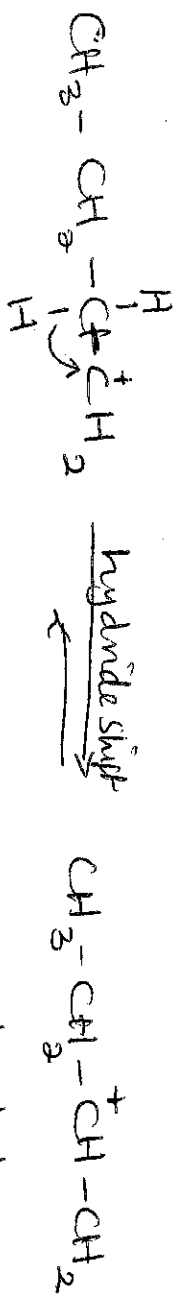
~~4~~ ④ Skeletal rearrangement of the alkyl group sometimes occurs during reaction, particularly when primary alkyl ~~halide~~ halides are used. Amount of rearrangement depends on catalyst, reaction temperature, and reaction solvents.

Less rearrangement is often found at lower reaction temperature but mixture of products are usually obtained.



—this is the result of carbocation rearrangement.

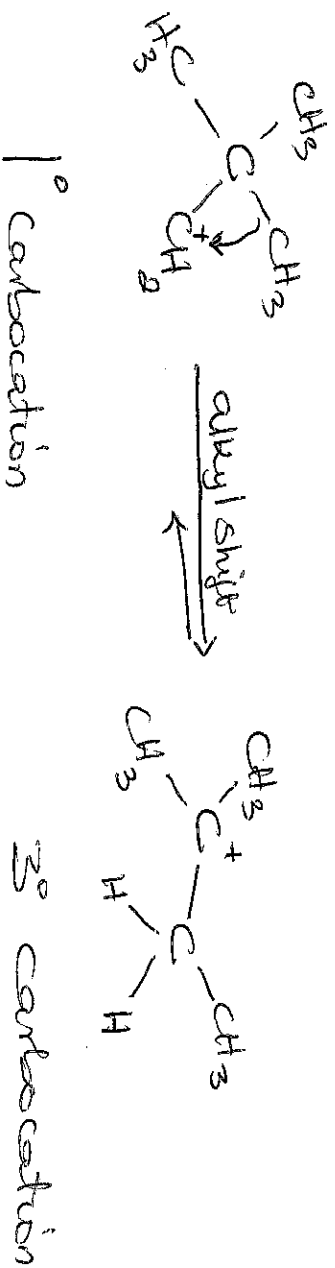
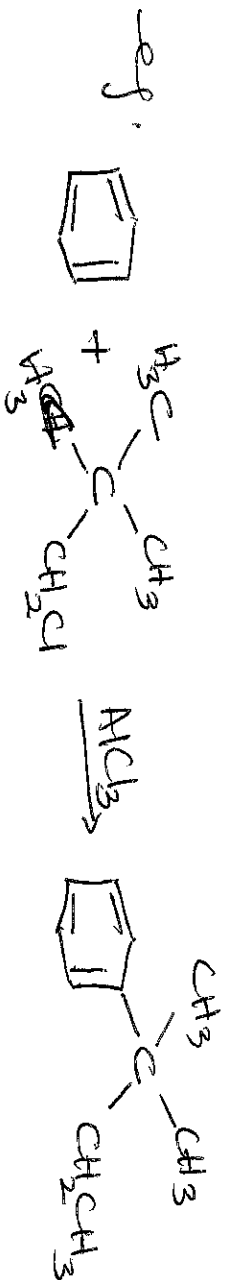
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Primary butyl-carbocation.

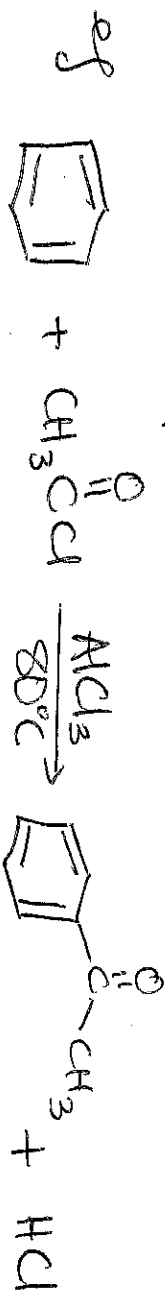
Sec-butyl carbocation

-rearrangement can also occur by alkyl shifts.



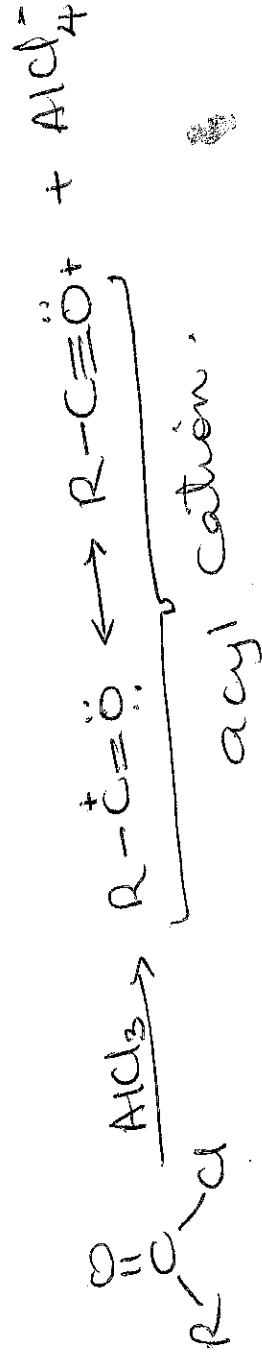
7. ACYLATION

- an aryl group -COR is introduced onto the benzene ring when an aromatic compound reacts with a carboxylic acid chloride, RCOCl, in the presence of AlCl₃.



mechanism is similar to that of Friedel-Crafts alkylation reactions:

The reactive electrophile is a resonance-stabilised ~~set~~ acyl cation, generated by reaction b/w the acyl chloride.



— Once formed the acyl cation does not rearrange but rather its attacked by an aromatic ring to give unarranged substitution product.

— acylation does not occur more than once in a ring b/c the product acylbenzene is always less reactive than the nonacylated starting material.

(8)

SUBSTITUENT EFFECTS IN SUBSTITUTED AROMATIC RINGS

Only one product can form when an electrophilic substitution occurs in benzene when an aromatic ring already has a substituent, the substituent already present on the ring has 2 effects.

① Substituents affect the reactivity of the ring. Some substituents activate the ring making it more reactive than benzene, and some deactivate the ring, making it less reactive than benzene.

In aromatic nitration, an $-OH$ group makes the ring 1000 times more reactive than benzene, while an $-NO_2$ substituent

makes the ring 2×10^7 times less reactive than benzene.

			
1000	1	0.033	6×10^{-8}
relative rate of nitration			

② Substituents affect orientation of the reaction. ~~3~~^{The} 3 possible disubstitution positions are Ortho, Para, meta are usually not formed in equal amounts. Instead the nature of the substituent present determines the position of the second substituent.

Some groups direct primarily to the Ortho and para positions while others to $\bar{e}$ meta.

(9)

Substituents can be divided
into 3 groups

(a) ortho, para-directing
activators

(b) ortho, para-directing
deactivators

(c) meta-directing deactivators.

No meta-directing activators are
known.

The directing effects of these
groups correlates with their
activities.

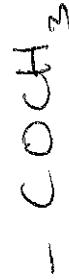
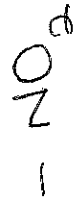
— all meta-directing groups are
strongly deactivating, most
para-directing groups are

activating.

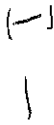
— Halogens are ~~sp²~~ ^{unconjugative},
ortho, para-directing but weakly

deactivating.

meta-directing deactivators



Ortho- and para-directing deactivators



Ortho- and para-directing activators



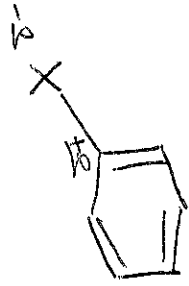
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Reactivity and orientation in aromatic electrophilic substitution are controlled by an interplay of inductive effects and resonance effects.

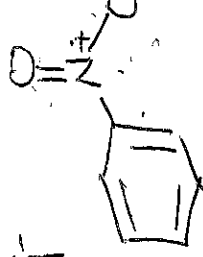
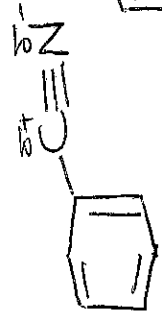
Inductive effect

— Simply the shifting of electrons in a bond, in response to the electronegativity of nearby atoms.

— σ bonds are due to the electrostatic attraction of atoms and the resultant polarity of bonds in functional groups. This results in withdrawal or donation of electrons through σ bonds (σ bonds). eg halogens, carbonyl groups, cyano groups and nitro group inductively withdraw electrons through the σ bond linking the substituent to the ring.



(X = F, Cl, Br, I)



Alkyl groups on the other hand, inductively donates electrons. The ~~reasons~~ for this behaviour involve the same factors that causes alkyl substituents to stabilize the alkenes & and carbocations. (Read stabilization of alkenes and carbocations by alkyl groups).

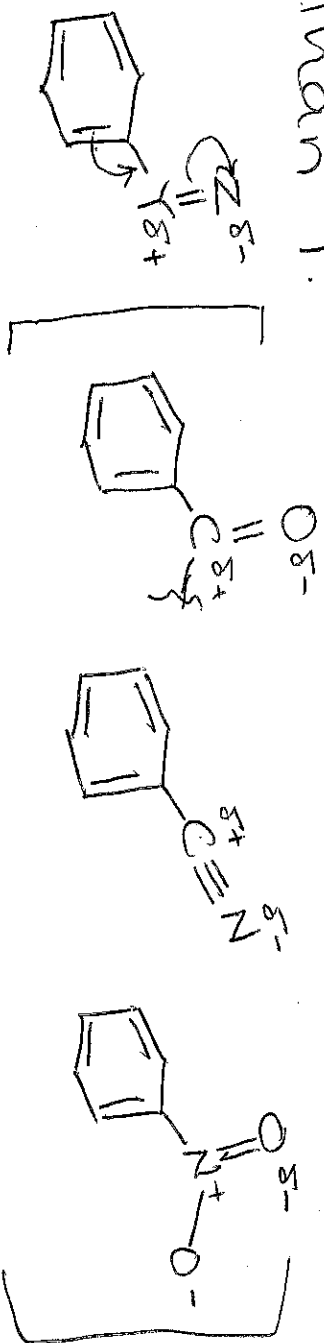


RESONANCE EFFECTS

Resonance effects are due to the overlap of a p-orbital on the substituent with a p-orbital on the aromatic ring and result

Note Substituents with the electron-withdrawing ^{resonance} effects have the

general structure $-Y=Z$, where the Z atom is more electronegative than Y.



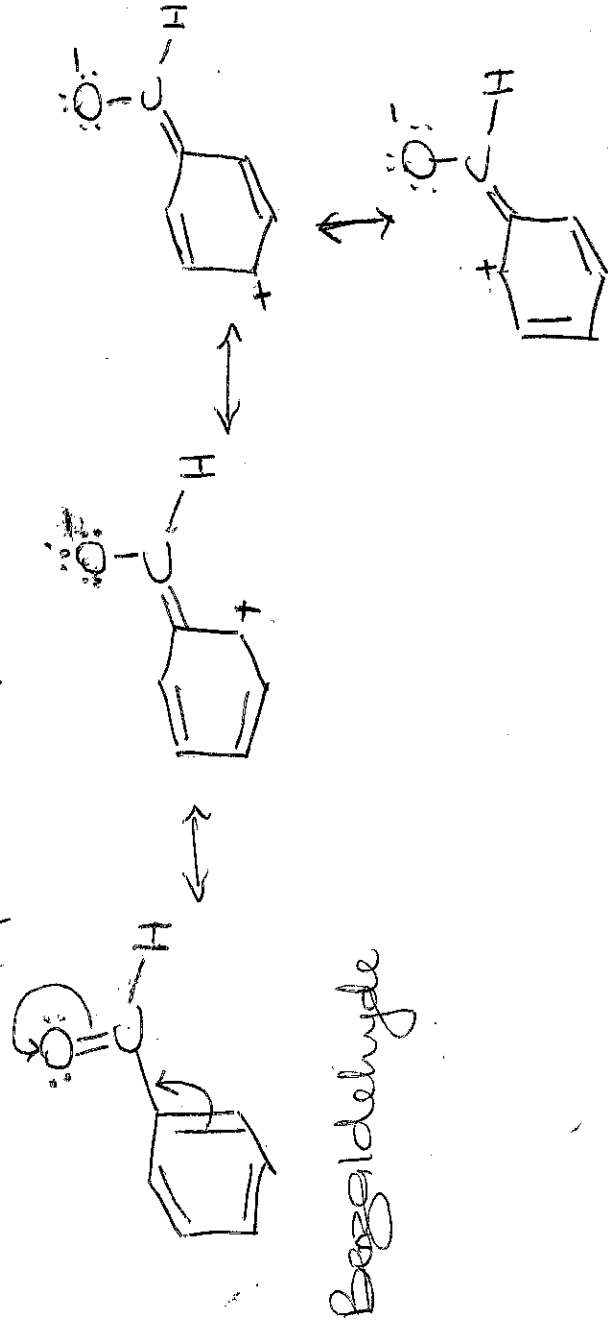
Halogens, hydroxyl, alkoxy ($-OR$) and amino substituents donate electrons to the aromatic ring by resonance. π electrons flow from the substituents to the ring, placing a $-ve$ charge in the ring as shown by the resonance structures for chlorobenzene. The effect is greatest at the

Ortho and Para positions as shown below.

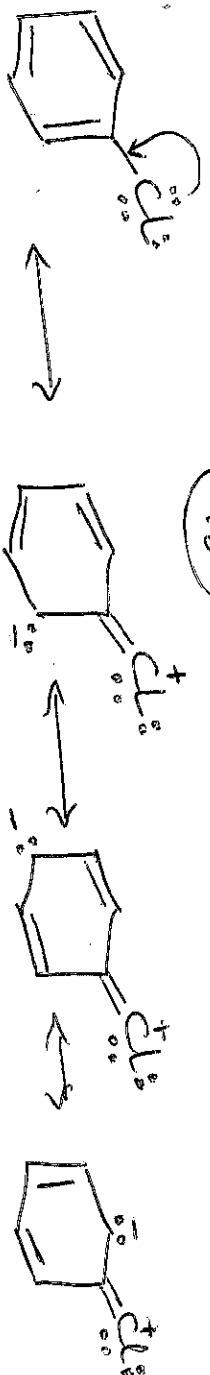
(11)

is withdrawal or donation of electrons through π bonds. Carbonyl, cyano, and nitrosubstituents, for example withdraw electrons from the aromatic ring by resonance. π electrons flow from the ring to the substituents placing a +ve charge in the ring.

of resonance structure of benzaldehyde. ~~with~~ The effect is greater at the ortho and para positions.

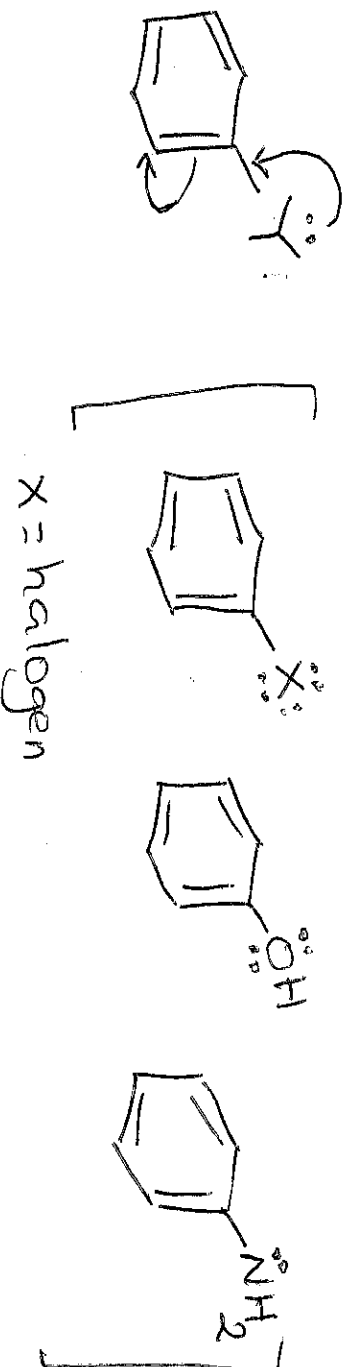


(12)



Chlorobenzene

Substituents with an electron donating resonance effect have the general structure $-Y$, where Y has a lone pair of electrons available for donation to the ring.



Note: It is not necessary for inductive effect and resonance effects to act in the same direction. Halogens, hydroxyl, alkoxy, and amino substituents for example, have electron withdrawing inductive effects

bles of the electronegativity of the atom bonded to the aromatic ring but have electron-donating resonance effects b/c of the lone-pair electrons on the atom bonded to the ring.

How does inductive and resonance effect result in the activation or deactivation of an aromatic ring towards electrophilic substitution.

The common feature of all activating group is that they donate electrons to the ring, thereby stabilizing the carbocation

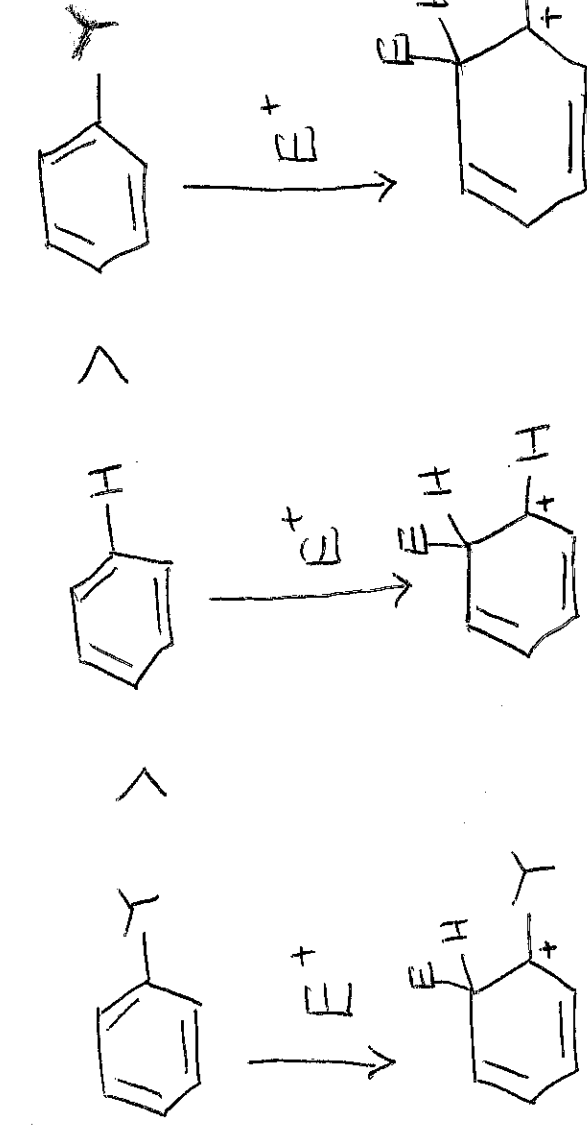
(13)

intermediate from electrophilic addition and causing it to form faster.

Hydroxyl, alkoxy, and amino groups are activating because their stronger electron-donating resonance effect outweighs their weaker electron-withdrawing inductive effect. Alkyl groups are activating b/cos ^{of} their electron-donating inductive effect.

The common feature of all deactivating groups is that they withdraw electrons from the ring, thereby destabilizing the carbocation intermediate and causing it to form more slowly.

Carbonyl, cyano, and nitro groups
 are deactivating because of
 both electron-withdrawing resonance
 and inductive effects. Halogens
 are deactivating because of their
 stronger electron-withdrawing
 inductive effect outweighs
 their weaker electron-donating
 resonance effect.



← reactivity

Y is an electron
 donor; Carbocation
 intermediate is
 more stabilized and
 ring is more ~~active~~ ^{reactive}.

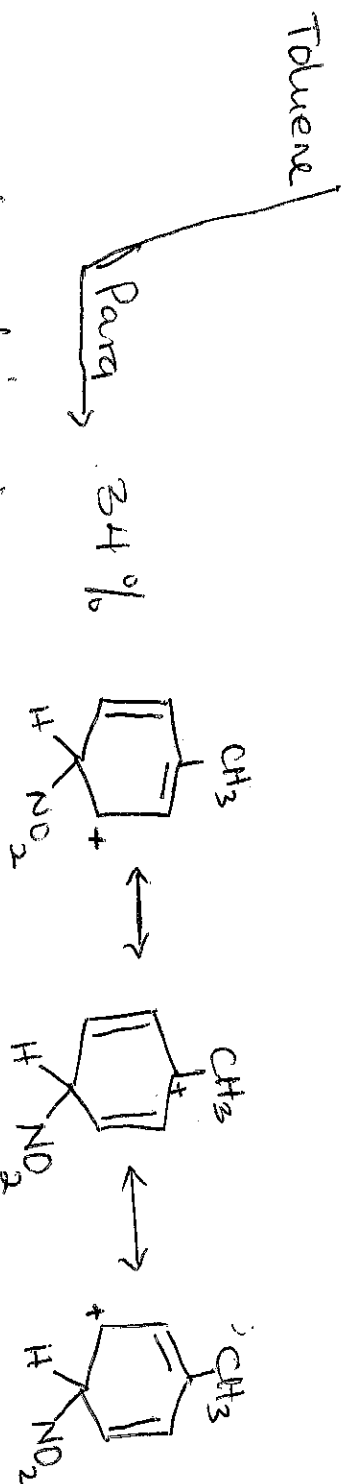
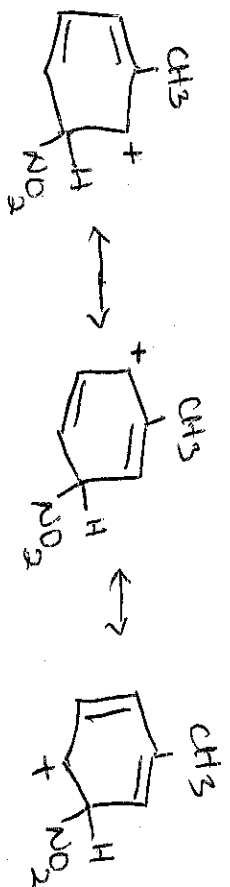
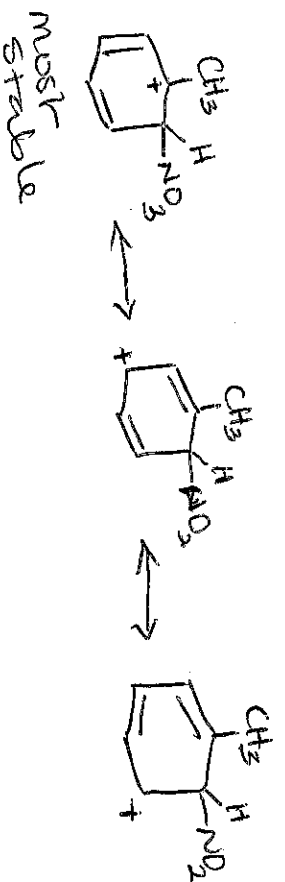
Y is an electron acceptor
 Carbocation intermediate
 is less stabilized,
 and ring is less
 reactive.

(14)

O, P-directing activation: alkyl

Inductive and resonance effects account for the directing ability of substituents as well as for their activating or deactivating ability. eg alkyl groups $\bar{e}n$ have an electron-donating inductive effect and behave as O, P directors.

See result of toluene nitration below:-



Carbocation intermediates in the rxn of toluene: Ortho and Para intermediates are more

Stable than meta intermediate.

Nitration of toluene might occur at either O, m or p to CH_3 group, giving the three carbocation intermediates as shown above. Although all 3 intermediates are resonance stabilized, the O, p intermediates are ^{most} ~~more~~ stabilized.

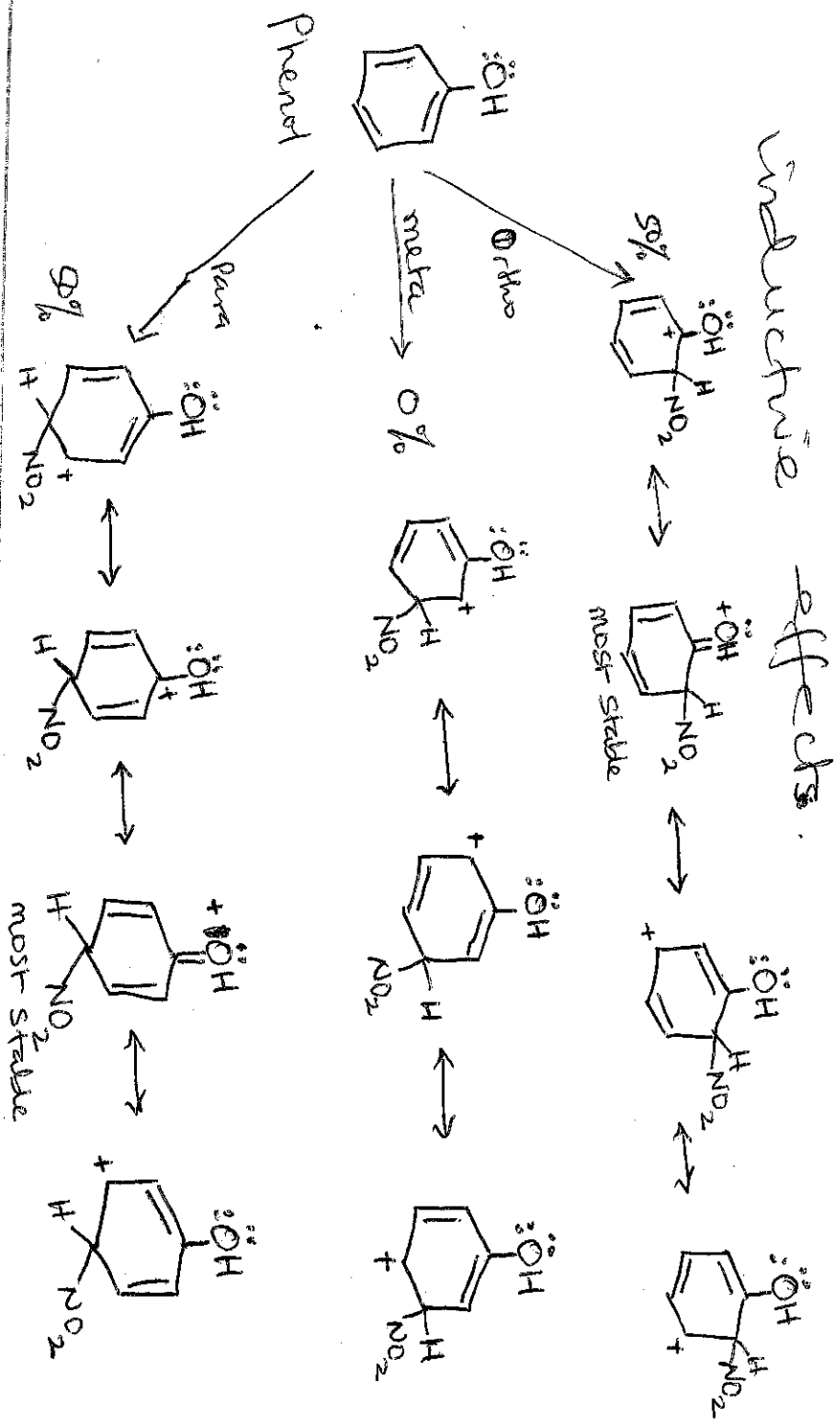
For O, p attack, a resonance form places the +ve charge directly on the methyl-substituted carbon, where it is in a 3° position and can best be stabilized by the electron donating inductive effect of the ~~CH~~ methyl group, but not so for meta.

The O, p intermediates are lower in energy than the meta intermediate and therefore form faster.

OH and NH₂.

Hydroxyl, alkoxy and amines
groups are O, P activators but for a different reason than alkyl grps.

Hydroxyl, alkoxy and amine have a strong, electron donating resonance effect that is most pronounced at the O, P positions and that outweighs a weaker electron-withdrawing inductive effects.



Carbocation intermediate in the nitration of ~~benzene~~ phenol. The Ortho and para intermediates are more stable than the meta intermediate b/c of π resonance donation of electrons from oxygen.

Halogens are O, P-directing deactivators b/c π strong electron-withdrawing inductive effect outweighs their $\sim$ weaker electron donating ~~inductive~~ resonance effects.

Though weak the electron donating resonance effects are felt only at the ortho and para positions. Thus halogen substituents can stabilise the +ve charge of the carbocation intermediate from ortho and para attack in the same way as -OH and -NH₂ substituents can. Meta intermediates

(16)

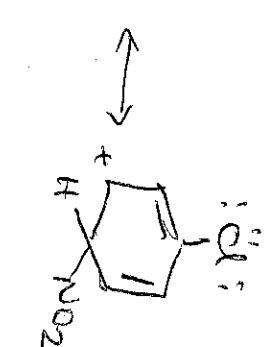
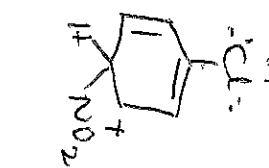
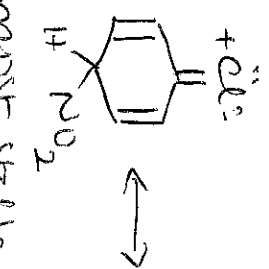
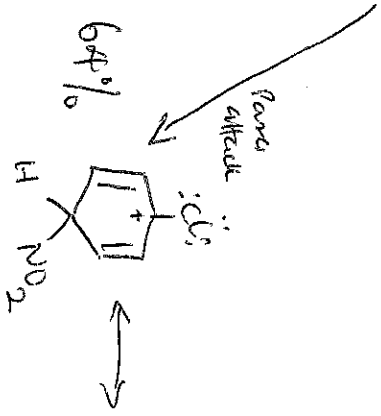
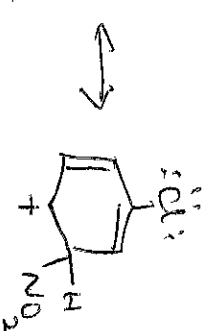
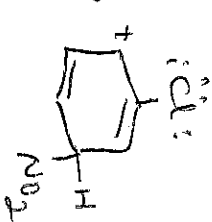
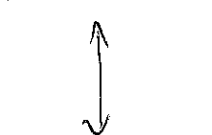
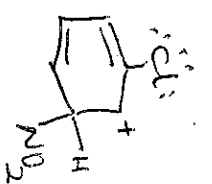
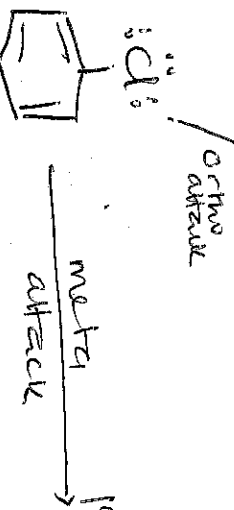
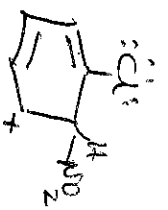
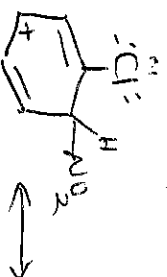
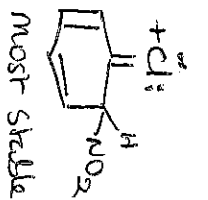
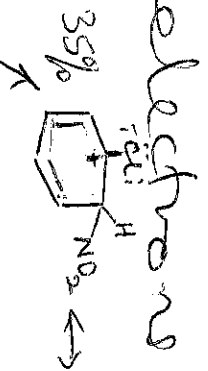
are not stabilized that way.

Note: Halogens, $-OH$, $-NH_2$, $-OR$ all with draw electrons inductively and donate e^- s by resonance.

Halogens are deactivators bc they have strong electron-withdrawing inductive effect and weak electron-donating resonance effect which is opposite to that of $-OH$, $-OR$, NH_2 etc

are activators. But all are o,p directors ~~except~~ ^{bc} lone pair of

electrons.



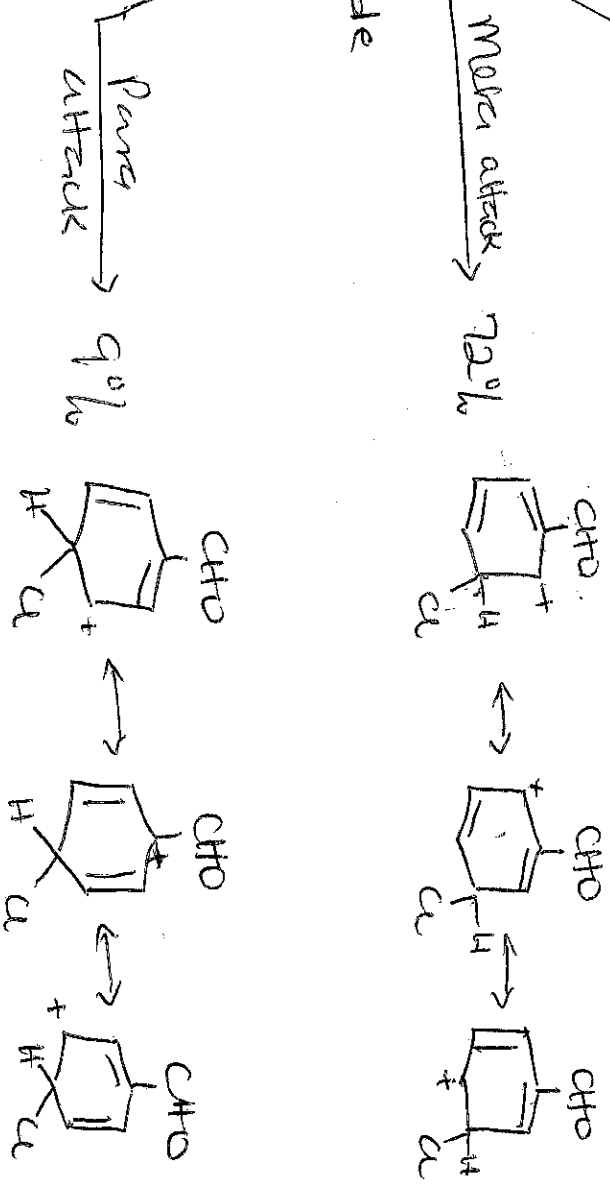
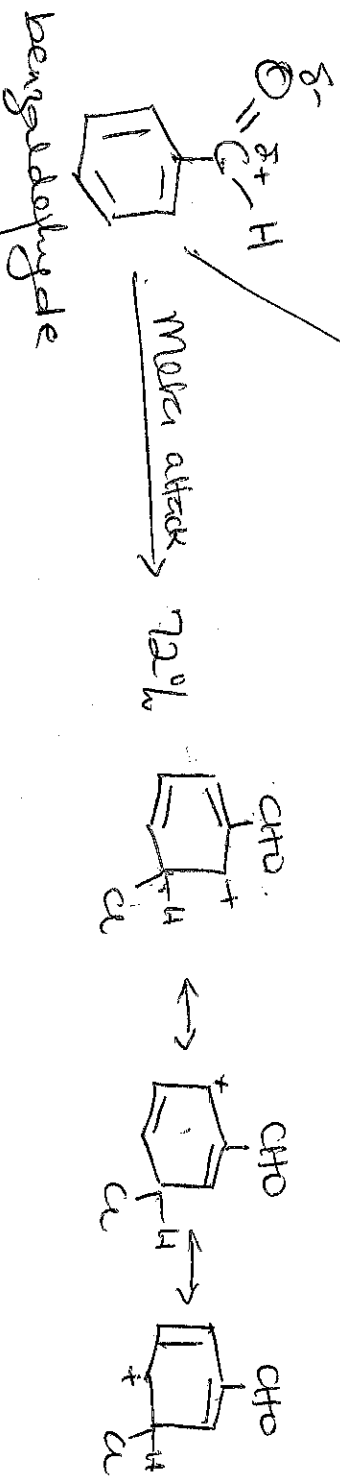
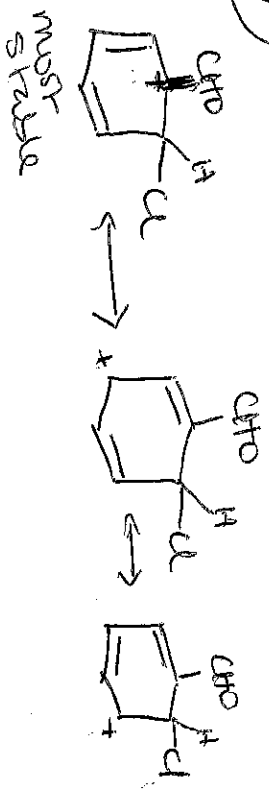
meta-directing deactivations

~~hydrogen~~ acts through a combination of inductive and resonance effects that reinforces each other.

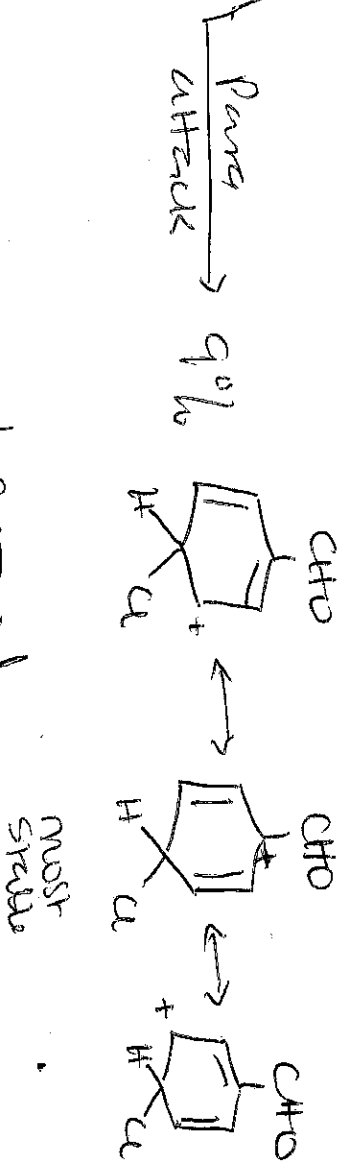
Inductively both ortho and para intermediates are destabilized b/c a resonance form places the true charge of the carbocation intermediate directly on the ring. Carbon atom that bears the deactivating group.

- resonance electron withdrawal is also felt at the o, p-positions thus reaction with an electrophile occurs at the meta position.

(17)



Chloroalkene of benzaldehyde. meta substitution.

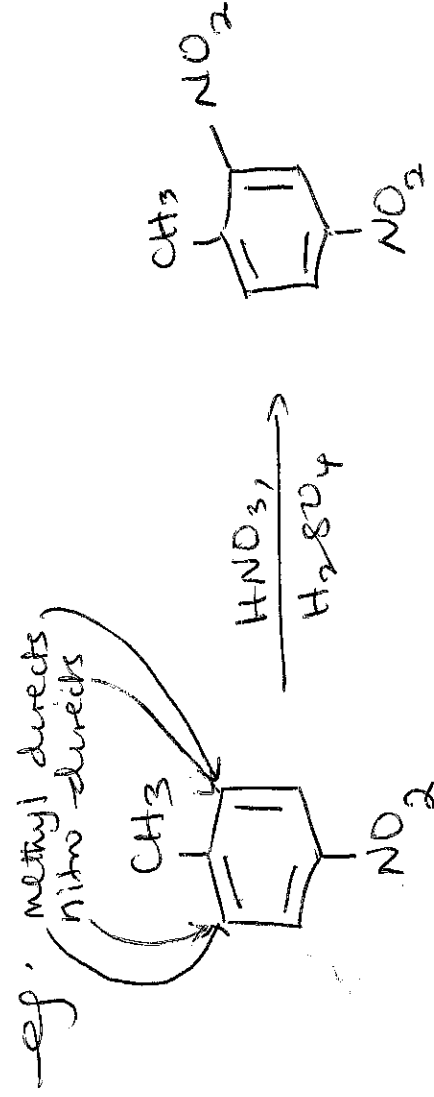


When there are already 2 substituents on the aromatic ring the position of the third group is governed by the same resonance and inductive effects. Now the additive

effects of 2 different groups will be taken into consideration.

In practice 3 rules help.

- ① If $\pm$ directing effects of the 2 grps reinforces each other there is no problem and one product is formed.

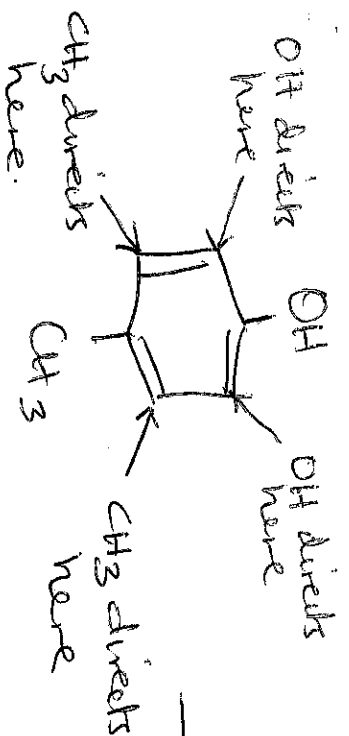


p-nitrotoluene

2,4-Dinitrotoluene

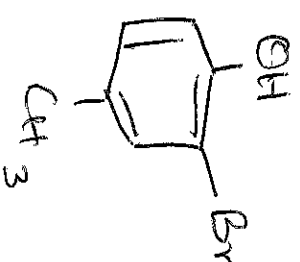
- ② If directing effects of $\pm$ 2 grps opposes each other, $\pm$ more powerful activating group has dominant influence, but mixtures of products often result.
eg

(18)



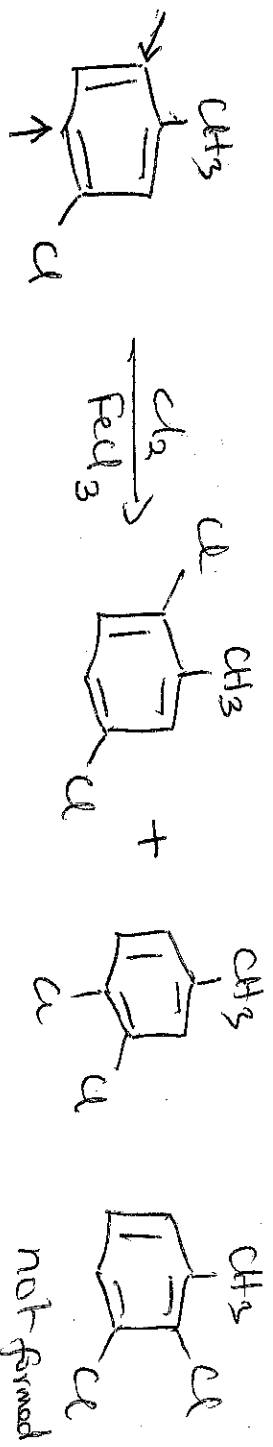
major product.

p-methyl phenol
(p-cresol)



③ In a meta disubstituted compd, further substitution rarely occur b/c 2 groups, b/c 2 site is too hindered.

eg



m-chloro toluene

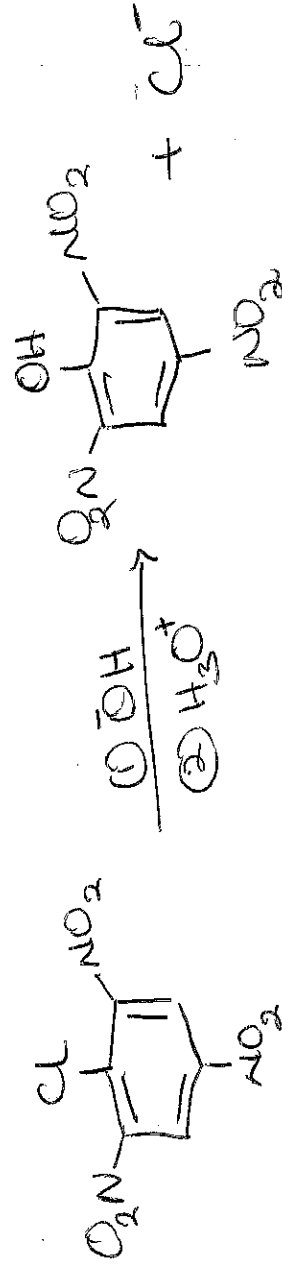
-Cl and -CH₃ are both o, p-directors.

Nucleophilic Aromatic Substitution

— usually occur by an electrophilic mechanism.

- Aryl halides have electron-withdrawing substituents but can also undergo nucleophilic aromatic substitution.

eg.



2,4,6-Trinitrochlorobenzene

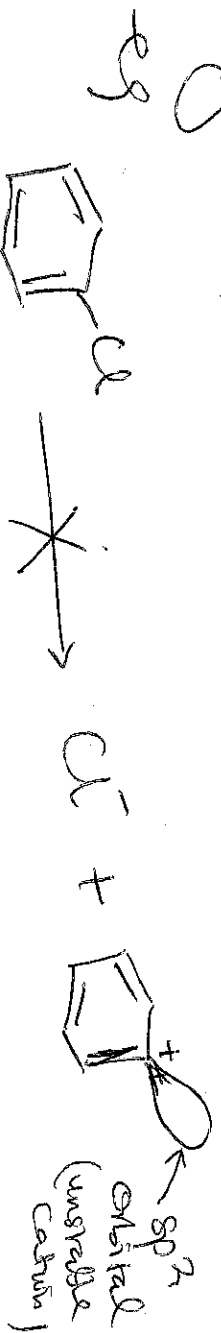
2,4,6-Trinitrophenol (100%)

The nucleophile OH^- substitutes Cl^- .

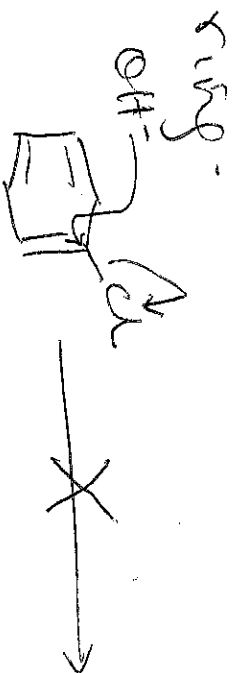
- ~~Similar~~ Resembles $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ rxns of alkyl halides but its diff b/c aryl halides are inert to $\text{S}_{\text{N}}1/\text{S}_{\text{N}}2$ rxns

(19)

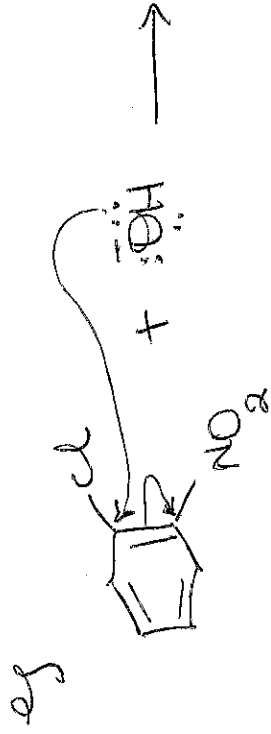
~~Ar~~ Aryl halides does not undergo S_N1 bc aryl cations are relatively unstable. Dissociation of aryl halides are energetically unfavourable.



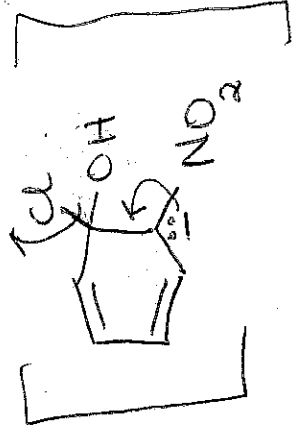
Aryl halide does not undergo S_N2 bc the halo-substituted carbon atom is sterically shielded from back-side attack by aromatic



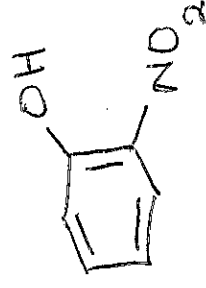
Nucleophilic aromatic substitution proceeds by the addition-elimination mechanism.



Nucleophilic addition of OH^- to π -electron-poor aromatic ring to yield a stabilized carbanion intermediate



The carbanion undergoes elimination of Cl^- to give a substituted product.



Above shows the mechanism of aromatic nucleophilic substitution. $\text{S}_{\text{N}}2$ occurs in 2 steps and involves a stabilized carbanion intermediate.

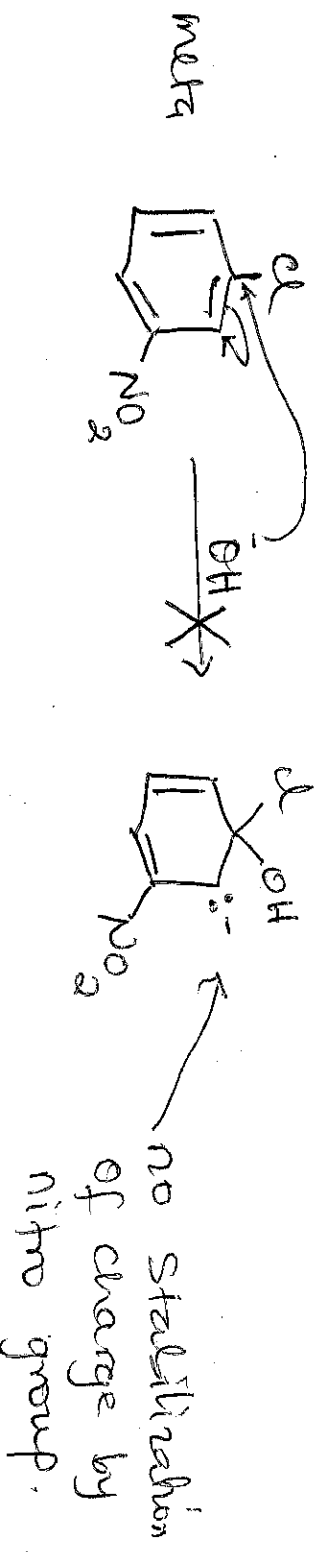
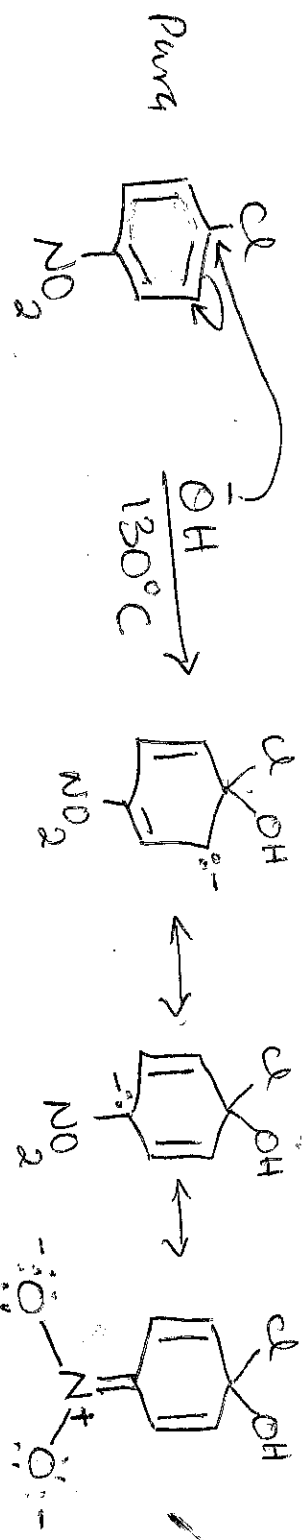
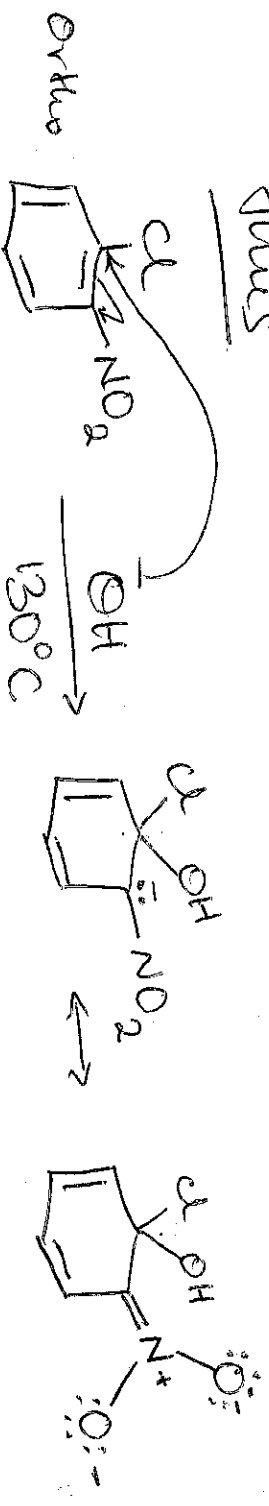
— Nucleophilic aromatic ~~substit~~ substitution occurs only when the ~~ring~~ aromatic ring has an e^- -withdrawing substituent in a position ~~o, or -p~~ to the halogen.

(20)

The more such substituents are $\bar{e}$ faster $\bar{e}$ rxn.

— only -O, $\bar{e}$ -p electron-withdrawing substituents can stabilize the anion intermediate through resonance; meta substituents will not give such resonance stabilization.

thus



Differences b/w electrophilic and
Nucleophilic aromatic substitution

- ① Electrophilic substitutions are favoured by electron-donating substituents, while nucleophilic substitutions are favoured by electron-withdrawing substituents.
- ② The electron-withdrawing groups that deactivate rings & electrophilic substitution (nitro, carbonyl, cyano, etc) activates them for nucleophilic substitutions.
- ③ These groups are meta-directors in electrophilic substitutions, but are ortho, para-directors in nucleophilic substitutions.

(21)

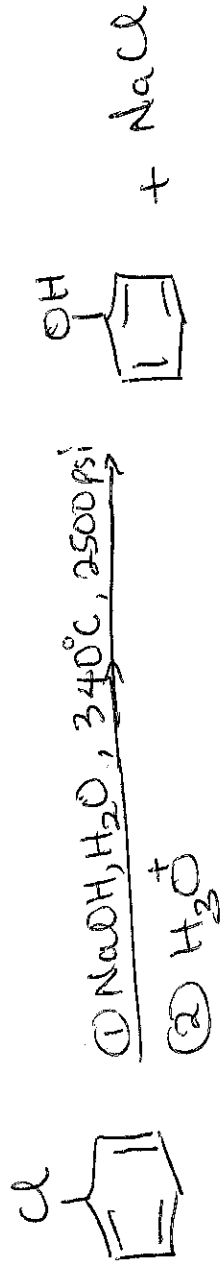
Nucleophilic aromatic substitution can also take place through a benzyne intermediate.

In this case it is elimination b/c additivity rather than the normal additivity b/c elimination.

— halo benzenes but electron-withdrawing ~~groups~~ substituents are inert to nucleophiles under most

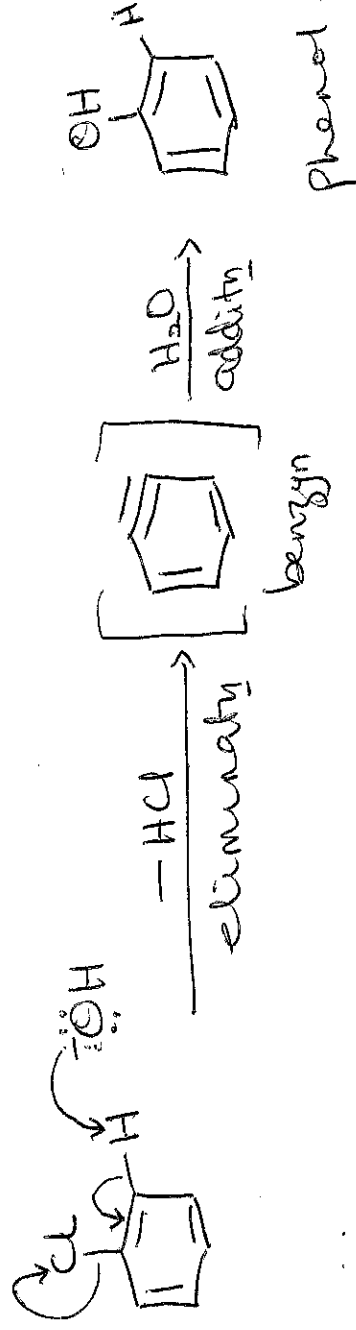
conditions. at high temp and pressures even chlorobenzene can be forced to react.

— Industrial preparation of phenol from chlorobenzene ~~with~~ by treatment of chlorobenzene with NaOH at 340°C under 3500 psi (pounds per sq. inch) pressure.

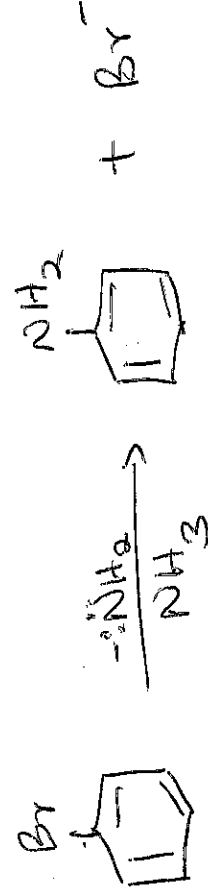


Mechanism

Strong base first eliminates HX from halobenzene in an E2 rxn, adds to benzyne in the 2nd step to give the product.



another example:

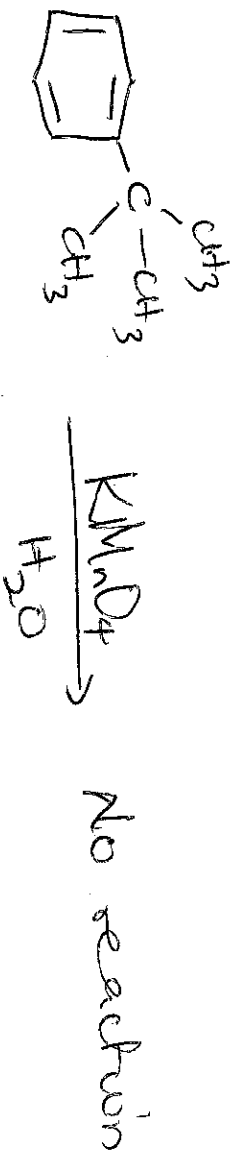
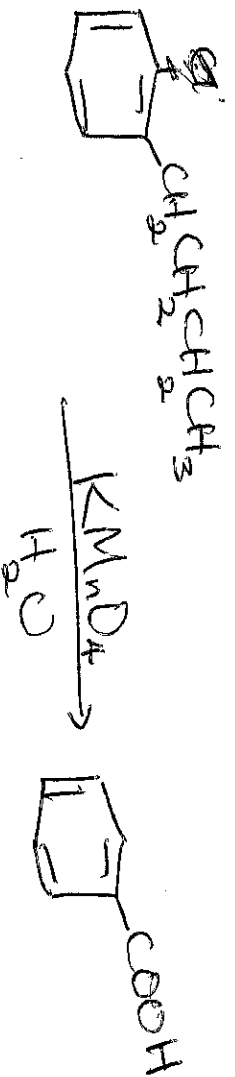
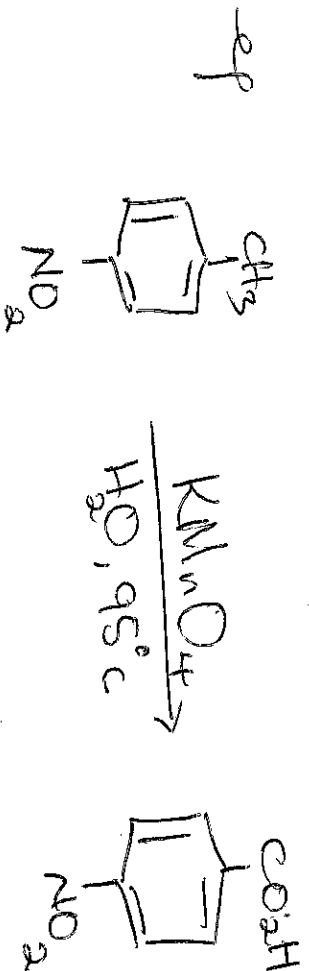


(22)

Oxidation of alkyl benzene side chains

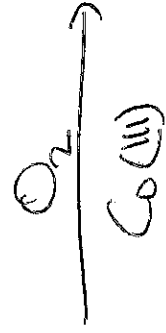
- Benzene ring is inert to oxidising agents like KMnO_4 and $\text{Na}_2\text{Cr}_2\text{O}_7$, reagents that will cleave alkene carbon-carbon bonds.

- alkyl side chains are readily oxidised.



Reason: attack is normally on the C-H bond at position next to π aromatic ring which is absent in t-butyl benzene

Industrially



Terephthalic acid
used in production of
polyester fibres.

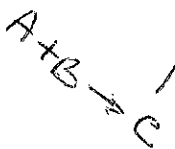
Types of Reactions in Organic Chemistry

- addition
- elimination
- substitution
- rearrangement

①

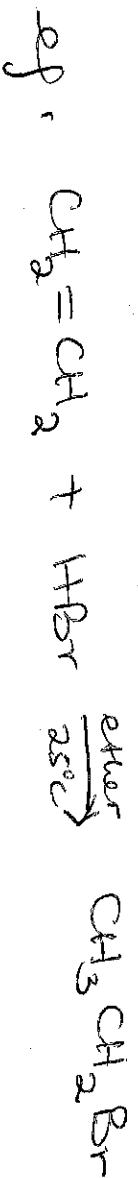
Addition Reactions

Two reactants to add together to form single new products



Organic compounds eg alkenes, alkynes etc.

- the ~~C=C~~ double bond is electron rich and can donate a pair of electrons
- rxns known as electrophilic addition rxns.



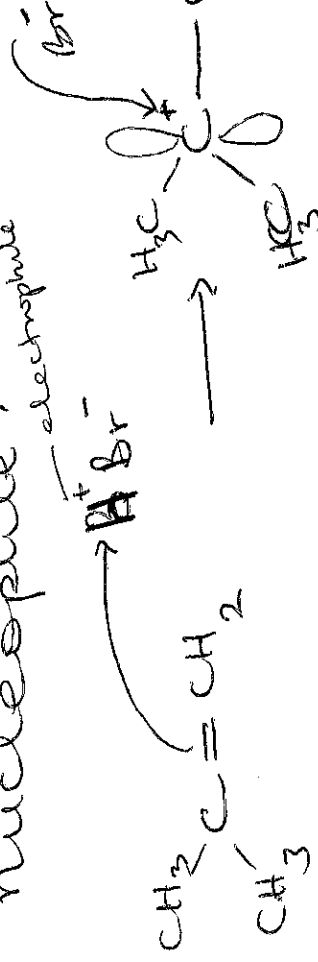
it can add HCl, HI etc.

take place in 2 steps

① formation of carbocation,

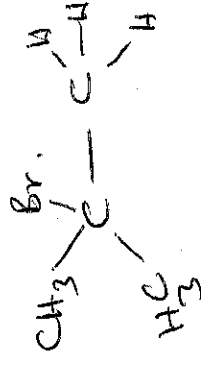
② carbocation reaction with Br^-

nucleophile



carbocation intermediate

Br^- donates an electron pair to relatively charged carbon atom.



Orientation of electrophilic addition

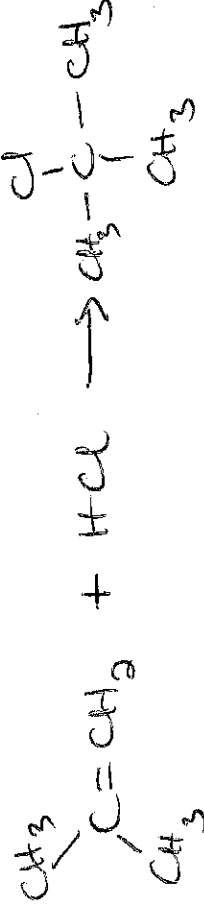
In case of unsymmetrical

substituted alkenes a single product is

formed instead of 2 that might be expected.

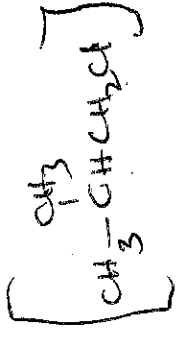
The reactions are

regiospecific.



CH_3

sole product



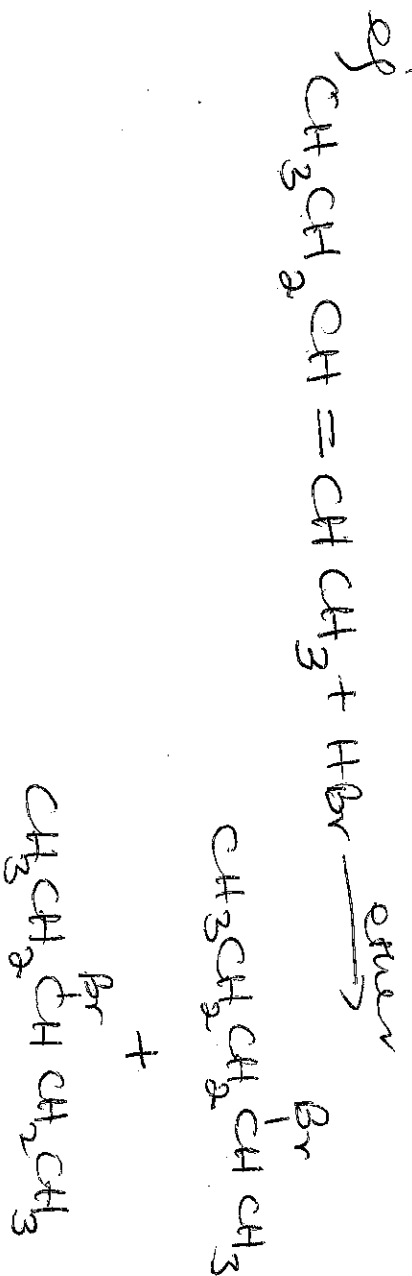
not formed.

(24)

Markovnikov's rule

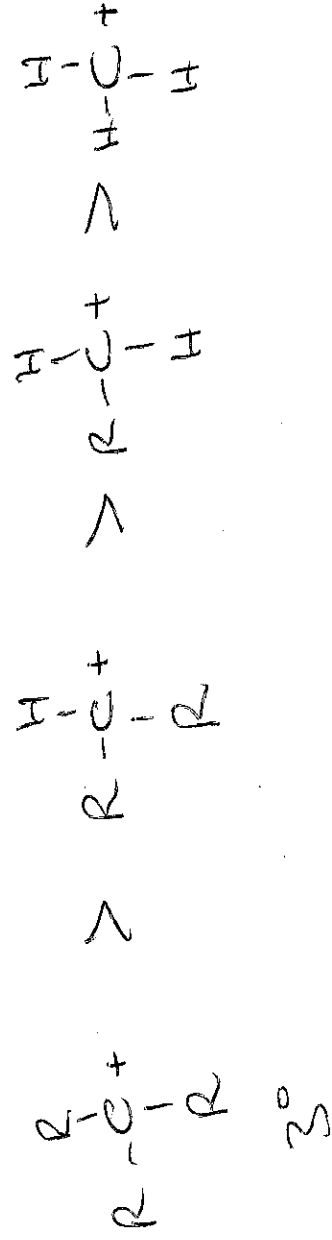
In the addition of HX to an alkene, the H attaches to the carbon with fewer alkyl substituents, and the X attaches to the carbon with more alkyl substituents.

Note: when both ends of double bond has the same degree of substitution, a mixture of products results.



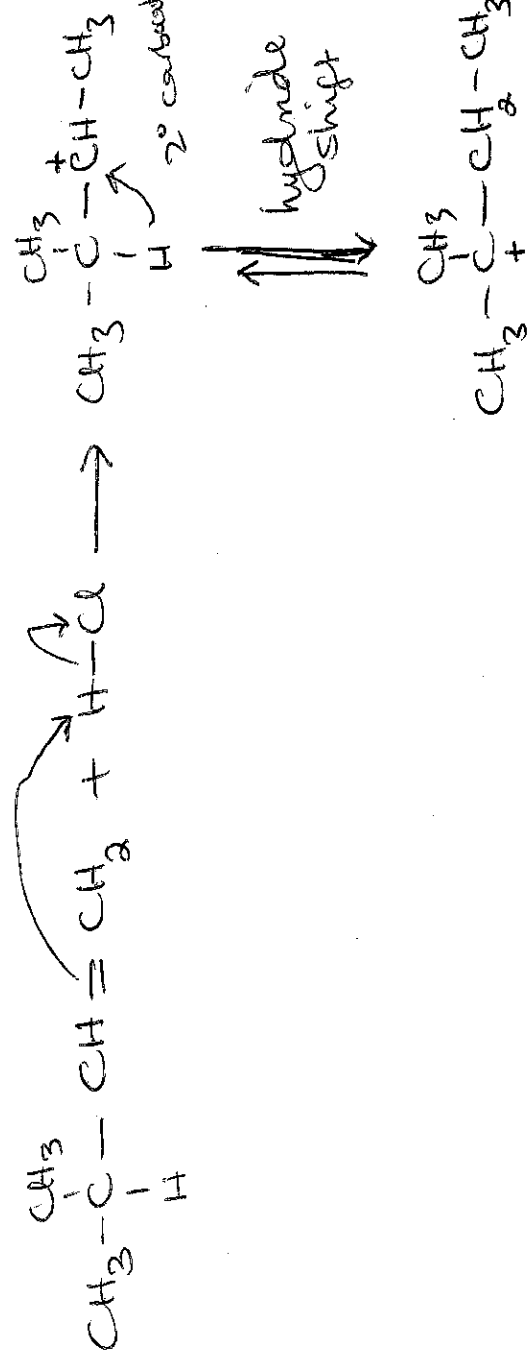
Since carbocation intermediates are involved, the Stability and more stable the

Structure of the carbocation plays a role.



more stable $\longleftarrow$ less stable

Carbocation Rearrangement



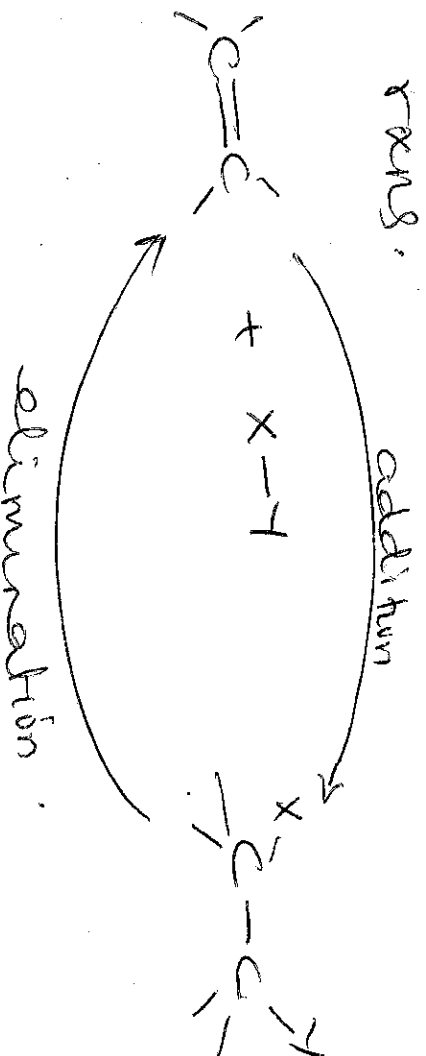
~~Products in this case will be~~
 Products in this case will be 50:50 of 2-chloro-3-methyl butane and 2-chloro-2-methyl butane.

Carbocation rearrangement can also occur by ~~alkyl~~ ^{alkyl} shift.

(25)

Elimination Reactions

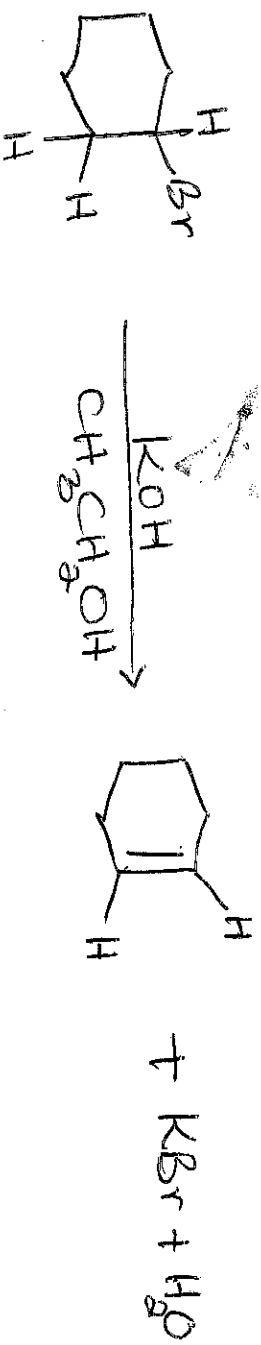
Preparation of alkenes are dominated by elimination



of ~~dehydrohalogenation~~
dehydrohalogenation and
dehydration.

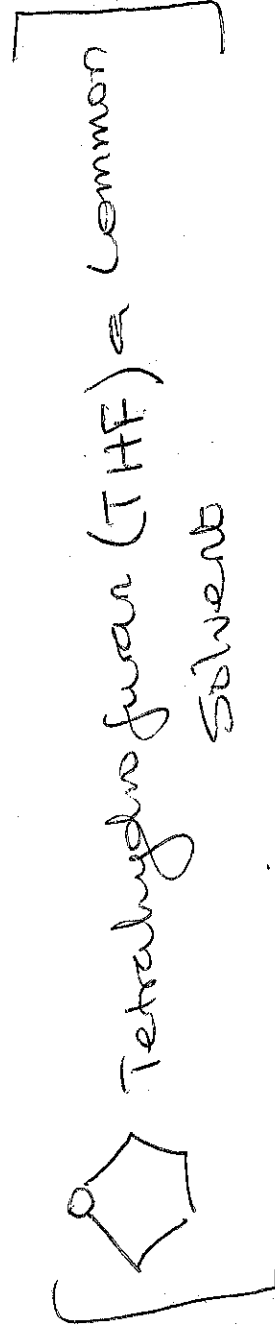
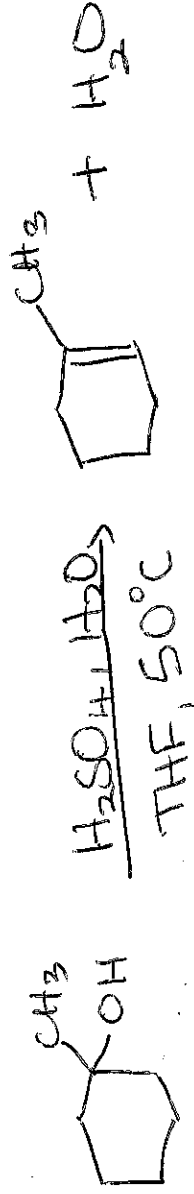
Dehydrohalogenation

occurs by rxn of an alkyl halide
with a strong base eg KOH in
ethanol

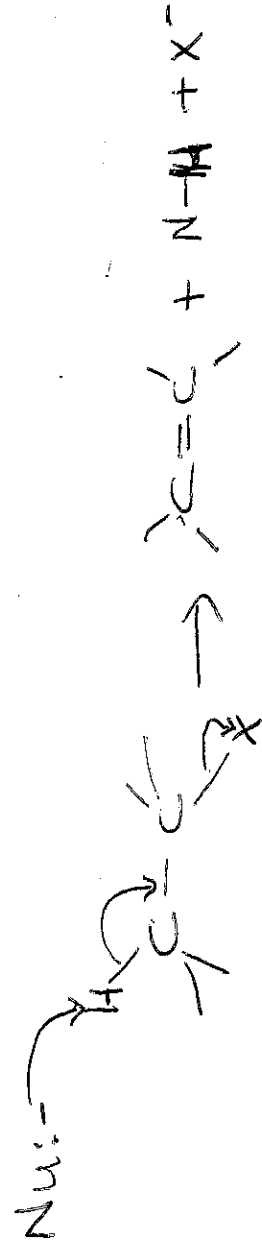


Dehydration:

By treatment of an alcohol with a strong acid.



Mechanism of elimination



Nucleophilic substitution and base induced elimination are two of the most widely occurring and versatile reactions in organic chemistry.

(26)

NUCLEOPHILIC SUBSTITUTION

Nucleophilic substitution reactions have to do with functional group transformations. Many different functional groups undergo this type of reactions. It involves an alkyl halide and a Lewis base that acts as the nucleophile. The substitution occurs at sp^3 hybridised carbons.

The substitution of an atom or group of atoms bonded to a carbon atom by a nucleophile is called nucleophilic substitution reaction.

In nucleophilic substitution reactions, the bond b/w the

sp^3 hybridized carbon and the leaving group are heterolytically broken - i.e. the leaving group leaves with the bonding electrons

- $\bar{e}$ leaving group refers to the atom or group of atoms that can be substituted by a nucleophile.

eg of leaving groups.

halogens; arene sulfonates, $ArSO_3^-$; ammonium ions, $-NR_3^+$; and alkyl sulfonium ions, $-SR_2^+$.

- The nucleophile is $\bar{e}$ reagents that substitute ^{for} the leaving groups. Many nucleophiles

are -vely charged - eg

$-OH^-$, CN^- , OR^- , $RCOO^-$, HS^- , N_3^-

Others are neutral molecules such as NH_3 , H_2O , alcohol.

(27)

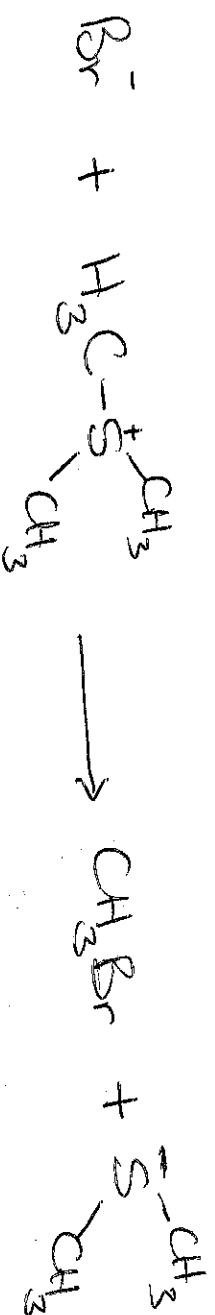
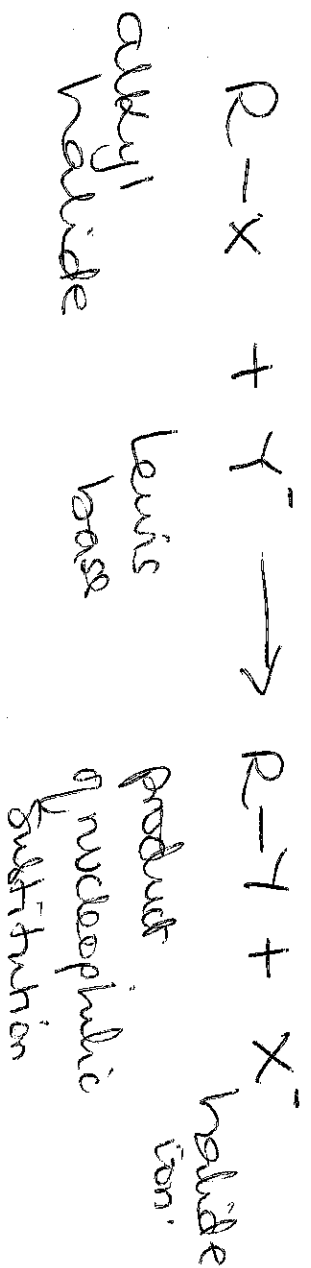
All nucleophiles have at least one pair of electrons to form a new bond with the sp^3 hybrid

Carbon atom.

~~new nucleophilic~~

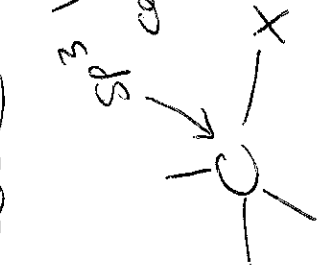
— ~~else~~ rxns are used extensively in organic synthesis.

examples

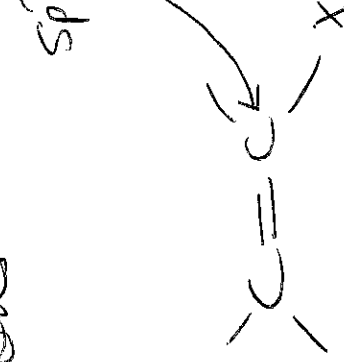


~~Nucleophilic~~

Alkyl halides are aryl halides, compounds in which halogen is attached to sp^2 hybridized carbons, are unreactive under these conditions.



alkyl halide



alkenyl halide



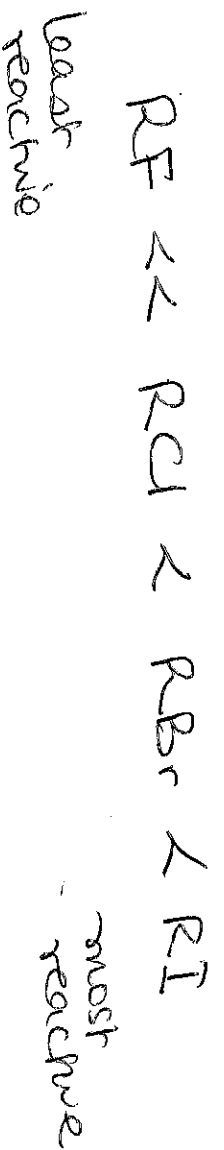
aryl halide

Choice of solvents is important to make sure reaction occurs in homogeneous solvents. many salts and alkyl halides are soluble in dimethyl sulfoxide (DMSO) - thus this makes a good medium for nucleophilic subst. rxns.

(28)

Nucleophilic Substitution reactions occur by two mechanisms: S_N1 and S_N2 mechanisms.

- Reactivity of halide leaving groups



Increasing rate of substitution by nucleophile.

- Same as their order in elimination rxn.

- Iodine has weakest bond to Carbon and iodide is $\bar{e}$ best

leaving group.

- Fluorine strongest bond and poorest leaving group.

- Alkyl fluorides rarely used as substrate in nucleophilic substitution.

reactions - to its ~~to~~ ~~reactive~~ several thousand times less reactive than chloride.

- leaving group ability is also related to basicity.

- a strongly basic anion is a poorer leaving group than a weakly basic one.

- Fluoride ion is the most basic and the poorest leaving group among $\bar{\text{I}}$ halide anions, while $\bar{\text{I}}$ is the best leaving group.

- In chemistry we speak often of rxns being fast or slow.

- exact rate at which rxn occurs is called rate of reaction or reaction rate and can be measured.

(29)

- factors that affect rxn rate
 - conc of reactants
 - temp
 - catalyst

- when rate is linearly dependent on conc of two reactants, the rxn is second-order reaction

~~when it depends on conc of only one of the reactants it is first-order reaction.~~

S_N2 Reaction Mechanism

- occurs with inversion of stereochemistry at the stereogenic carbon.
- shows second-order kinetics and follows the rate law

$$\text{Rate} = k \times [RX] \times [Nu^-]$$

S_N2 - means substitution, nucleophilic, bimolecular.

- Bimolecular means two molecules, nucleophile and alkyl halide; takes part in the step whose kinetics are measured.
- Kinetics - measure of relationship b/w reaction rate and reaction conditions.

* Features of S_N2 reaction.

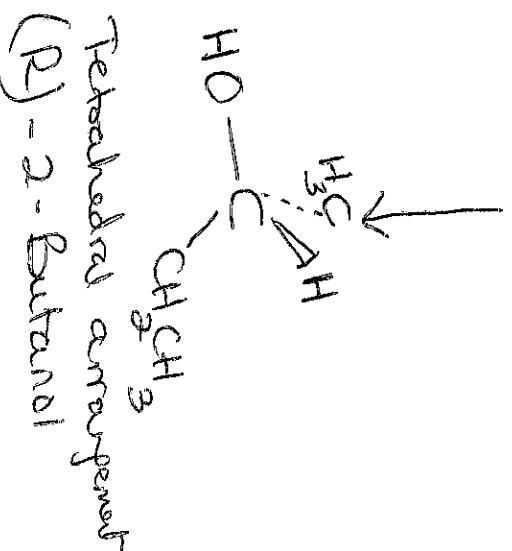
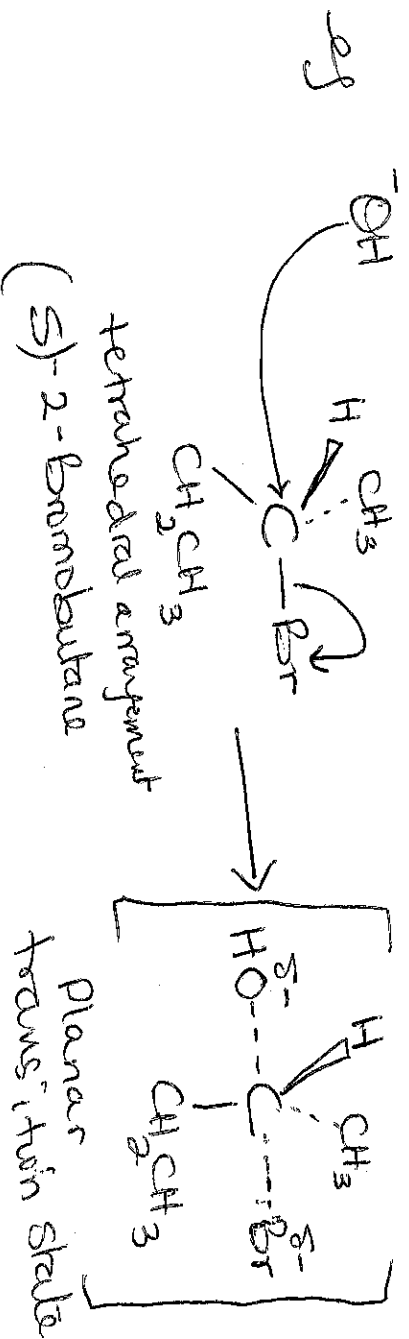
- rxn takes place in a single step — without intermediate
- backside attack
i.e. nucleophile attacks at a direction opposite to leaving group leading to inversion of configuration.

- ~~the~~ Occurs through a transition state in which new Nu-C bond is partially formed ~~at~~ the same time that the old C-Y bond is partially breaking — and $\bar{e}$ -ve charge is shared by both the

(3D)

the incoming nucleophile and the outgoing leaving group.

- remaining 3 bonds to carbon in the transition state are in a planar ~~area~~ arrangement.



- 2 reagents are involved in the step whose rate is measured.
- rate directly proportional to conc of both alkyl halide and nucleophile. It is first order wrt each reactant and overall second order

$$\text{Rate} = k[\text{RX}][\text{Nu}^-]$$

- rate determining step is bimolecular.

thus $\text{S}_{\text{N}}2$.

- inversion of configuration is often referred to as Walden inversion.

~~after a German chemist~~

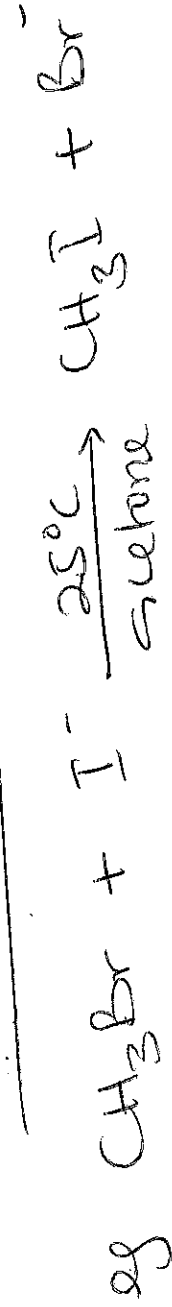
Factors that influence the rate of $\text{S}_{\text{N}}2$ reactions

(substrate)

Effect of alkyl group

(steric effect)

Structure



In both rxns above, rxn temp, solvent (acetone), the leaving group (Br^-) and the nucleophile

(31)

(I) are identical. The only differences are the ~~structures of the~~ bromoalkane and products.

Rate of rxn observed

CH_3Br	relative rate 145	Time at 1/2 of starting bromoalkane to react 1.65 mins
------------------------	----------------------	---

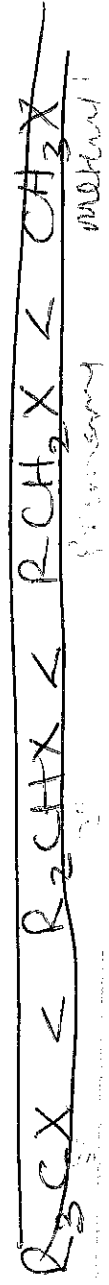
$\text{CH}_3\text{CH}_2\text{Br}$	1	240 mins (4 hrs)
-----------------------------------	---	------------------

See table

Bromo alkane	Relative rate	Time at 1/2 starting bromoalkane to react. mins	Hours	Days
CH_3Br	145	1.65	-	-
$\text{CH}_3\text{CH}_2\text{Br}$	1.0	240	4	-
$\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$	0.82	300	5	-
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$	7.8×10^{-3}	3.1×10^4	513	21
$\text{CH}_3\text{CH}(\text{CH}_3)\text{Br}$	2°	1.3×10^{-5}	1.8×10^7	3.1×10^5 (35)
$(\text{CH}_3)_3\text{CCH}_2\text{Br}$	1°	1.3×10^{-5}	1.8×10^7	3.1×10^5 (35)
$(\text{CH}_3)_3\text{CC-Br}$	negligible	-	-	-

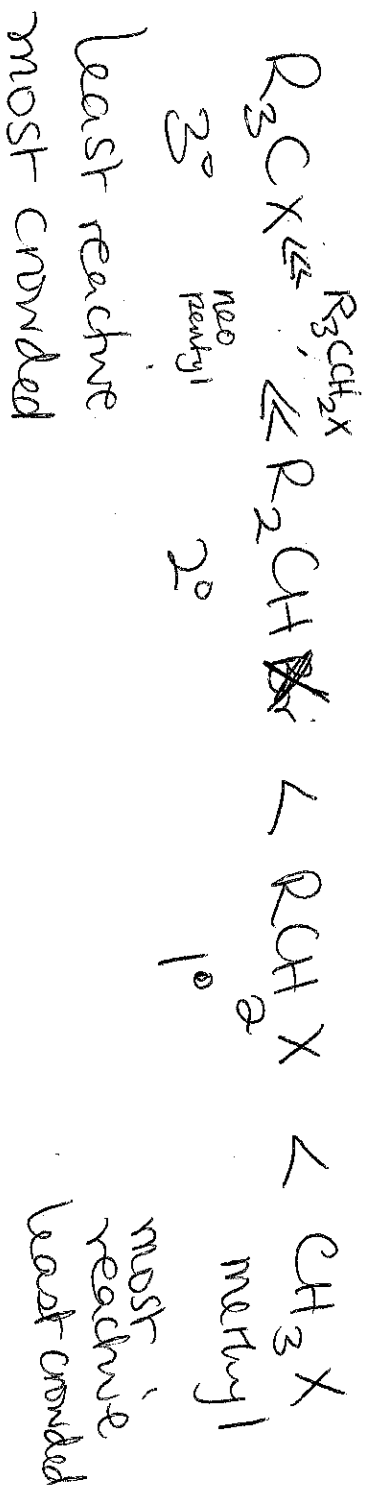
The ~~large~~ rate $\frac{d}{dt}$ b/w reaction
of series
of alkyl bromide and methyl
bromide is 10^6 when comparing
the most reactive member of
the series ~~at~~ methyl bromide
and the least reactive tert-
butyl bromide.

— reason is steric hindrance—
each offers to nucleophilic
attack—since nucleophile
approaches from the backside.
 S_N2 reaction exhibits the
following dependence of rate
on substrate.



(3a)

Increasing rate of substitution
by S_N2 mechanism $\rightarrow$



alkyl group at carbon atom
adjacent to the point of nucleophilic
attack also decreases the rate of
the S_N2 reactions.

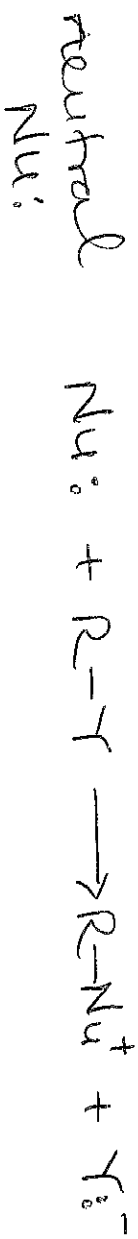
Note
When rxn center is CH_3X , S_N2 rxn
occurs rapidly, more slowly at a
2° carbon atom, very slowly at
a neo-pentyl carbon and not
at all at a 3° carbon atom,
but the rxn occurs by another
mechanism S_N1 .

(2) Nature of the attacking nucleophile (Nucleophile and nucleophilicity)

— A nucleophile ^{from rxn center} displaces a leaving group in S_N2 rxn. This step is the rate determining step of the rxn, this means the rate of S_N2 rxn will vary as the nucleophile is changed.

The nucleophile (Lewis base) can be -vely charged or neutral. When the nucleophile is -vely charged the product will be neutral but if the nucleophile is neutral product is +vely charged.

(33)

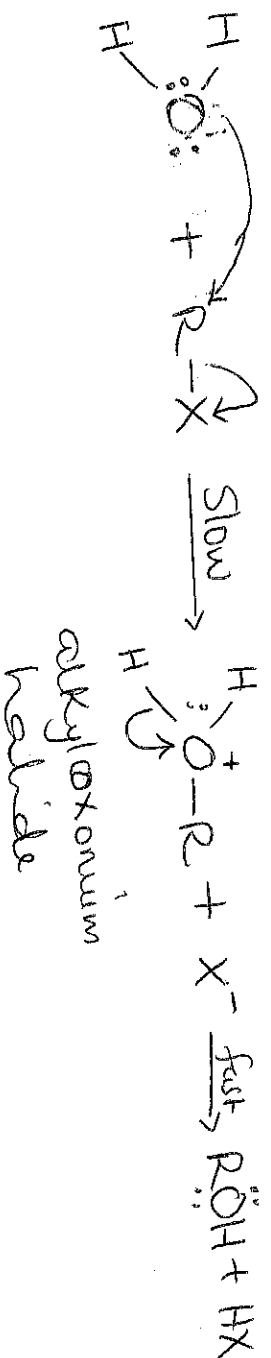


SOLVOLYSIS

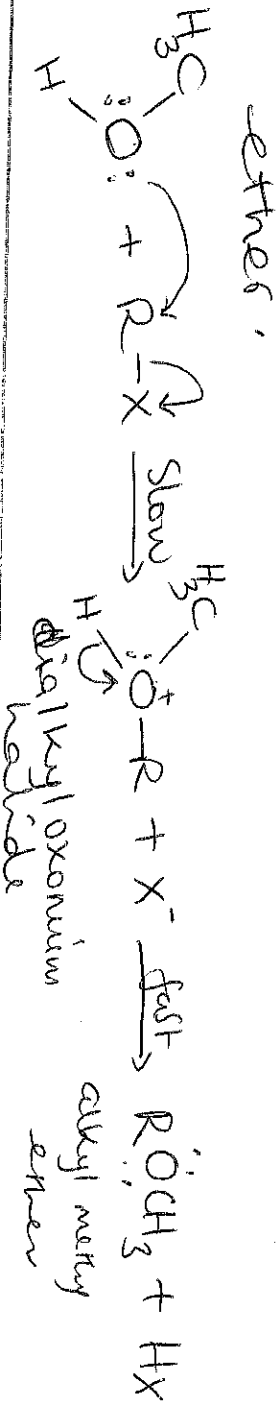
an eg of substitution rxn involving a neutral nucleophile.

in this reactions $\bar{\text{e}}$ nucleophile is $\bar{\text{e}}$ solvent in which $\bar{\text{e}}$ rxn is carried out.

eg solvolysis in water converts alkyl halide to alcohol.



Solvolysis in methyl alcohol converts alkyl halide to alkyl methyl ether.



Nucleophilicity:-

is a measure of how fast a nucleophile (Lewis base) displaces a leaving group from a suitable substrate.

Nucleophiles react at different rates.

A negatively charged nucleophile is more

reactive than a neutral one.

e.g. RO^- , RCOO^- more nucleophilic than ROH & RCOOH

~~Any~~ Any species, either negatively charged

or neutral can act as a

nucleophile provided it has a

lone pair of electrons.

To determine the nucleophil-

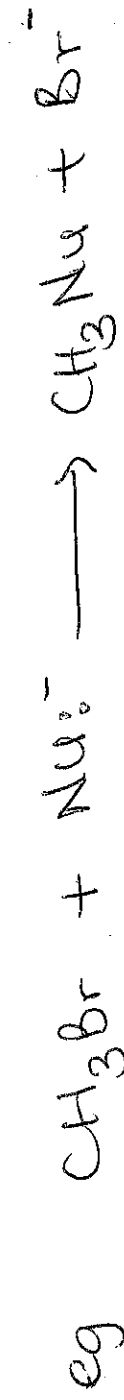
city of nucleophiles, the

different nucleophiles are used

under the same conditions of

substrate, solvent and reactant

concentrations.



(34)

Nu =	HS ⁻	CN ⁻	I ⁻	CH ₃ O ⁻	HO ⁻
relative reactivity	125,000	125,000	100,000	25,000	16,000

more reactive ← relative reactivity as nucleophile

Cl ⁻	NH ₃	H ₂ O
1000	700	1

← less reactive

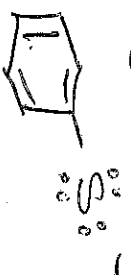
Some general trends that can be observed

⊕ Nucleophilicity increases with increasing base is a stronger nucleophile than an acid.

eg Acid

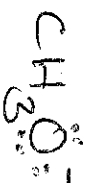
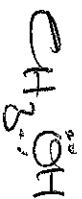


conjugate base



relative rate

21



83

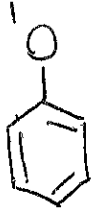
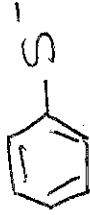
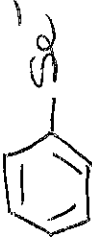
relative rate

0.000043

increasing nucleophilicity →

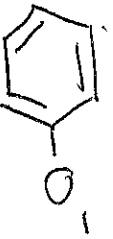
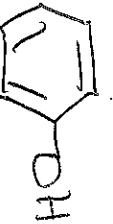
⑥ Nucleophilicity usually increases going down a column of the periodic table.

i.e. the nucleophilicity of ~~an~~ either anions or neutral compounds whose nucleophilicity atoms are in the same column of the periodic table generally increases going down the column.

Nucleophile	Relative Rate
F^-	0.017
Cl^-	1.0
Br^-	27
I^-	1,130
	24
	357,000
	2,333,000
$\ddot{S}(CH_3)_2$	15
$\ddot{Se}(CH_3)_2$	89

(35)

③ Nucleophilicity and basicity are generally related as long as the nucleophilic atom is the same

Nucleophile	Relative Rate of Reaction in iodomethane	Conjugate acid	pK _a of conjugate acid
CH_3O^-	83	CH_3OH	15.7
	24		9.89
$\text{CH}_3\text{C}(=\text{O})\text{O}^-$	0.9	$\text{CH}_3\text{C}(=\text{O})\text{OH}$	4.75

increasing nucleophilicity

— nucleophilic atom in each case is -ve oxygen atom and the stronger the base, the faster the substitution rxn.

Note. Nucleophilicity order can change depending on solvent.

Nucleophilicity is also related to polarizability, or ease of distortion of the electronic "cloud" around the nucleophile. Among related atoms, polarizability increases with increase in size. Thus iodide is more polarizable and most nucleophilic halide ion, fluoride is the least.

③ Nature of the leaving group.

b/c the leaving group is expelled with a -ve charge, in most S_N2 reactions, it is expected that the best leaving groups are those that best stabilize the -ve charge.

Furthermore, b/c $\bar{e}$ stability of an anion is related

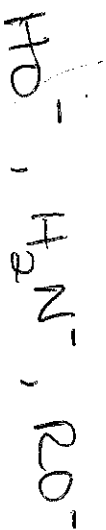
(26)

to its basicity, the best leaving group should be the weakest bases.

Weak bases (conjugates of strong acids) are the best leaving groups.

p-Toluenesulfonate (tosylate) leaving group is very easily displaced, as are iodides and bromide ions, but chloride and fluoride ions are much less effective as leaving groups.

Nucleophile	TosD ⁻	I ⁻	Br ⁻	Cl ⁻	F ⁻
Relative reactivity	60,000	30,000	10,000	200	1
more reactive	← reactivity as leaving grp → less reactive				

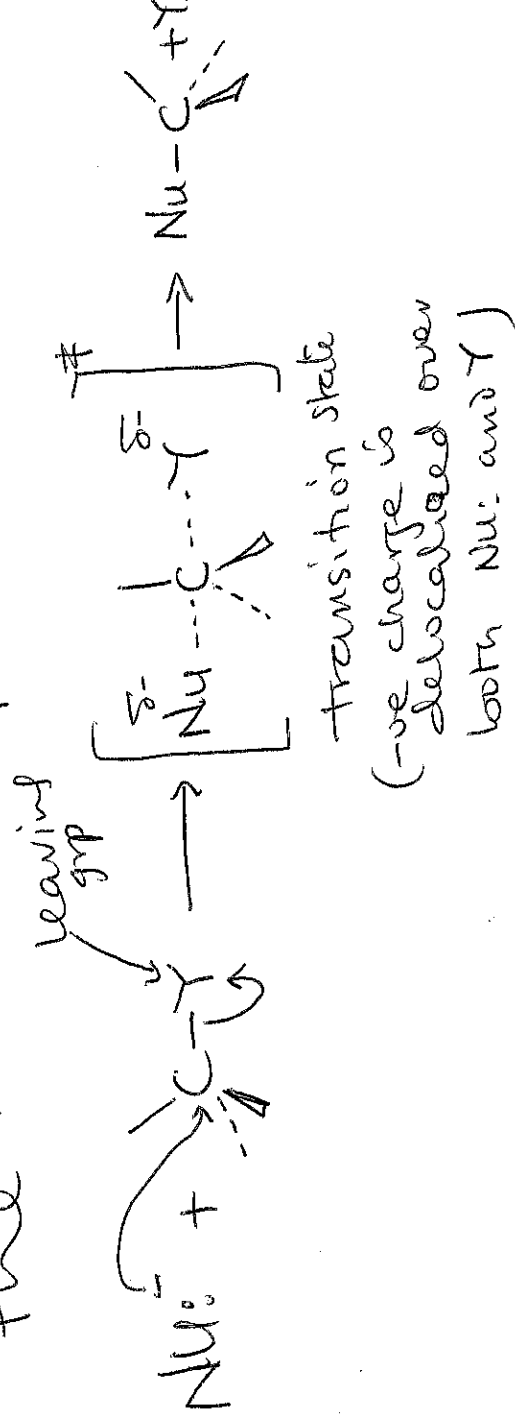


~0

Reasons why stable anions make good leaving groups.

- the transition state.

~~here the~~ In the transition state for S_N2 reaction, the charge is distributed over attacking nucleophile and leaving group. The greater the extent of charge distribution by leaving group, the lower the energy of the transition state and the more rapid the reaction.



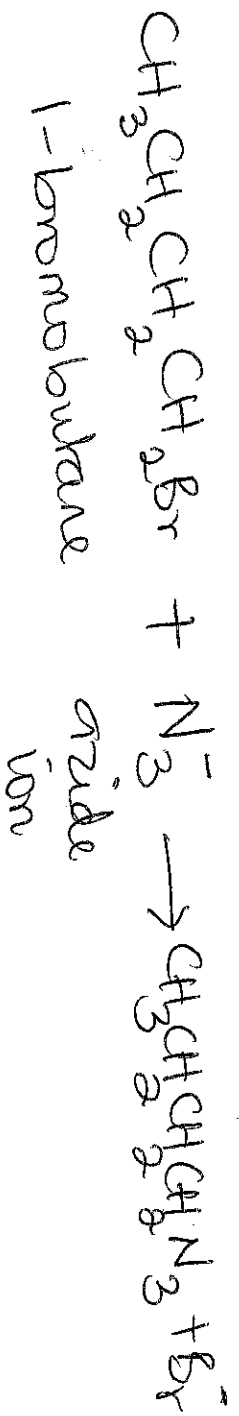
(27)

$R-F$, $R-OH$, $R-OR$, $R-NH_2$ do not undergo S_N2 reactions.

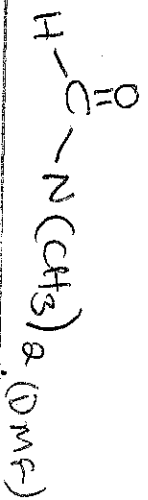
F^- , OH^- , OR^- , NH_2^- are very poor leaving groups.

④ The role of Solvents

A change in solvent strongly affects the rate of S_N2 reaction. eg changing the solvent from methanol to N,N -dimethylformamide (DMF) increases rate of rxn of isobutane and azide ion by a factor 2700.



Solvent	Relative rate
CH_3OH	1.0
$H-C(=O)-N(CH_3)_2$ (DMF)	2,700



Reason: B/c of diff abilities of solvents to solvate the reagents and transition states.

Solvation refers to the specific interactions b/w solvent molecules and dissolved reagents and/or transition state. These interactions are hydrogen bonding, dipole-dipole, and ion-dipole interactions.

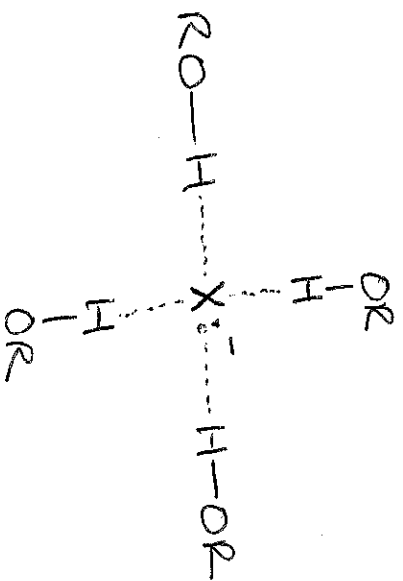
- Protic solvents which contain

- OH or -NH groups are generally worst solvents for S_N2 reactions; polar aprotic solvents, which have strong dipole but don't have -OH or -NH grps are the best.

- Protic solvents eg methanol and ethanol, slow down S_N2 rxns by clustering around the reacting

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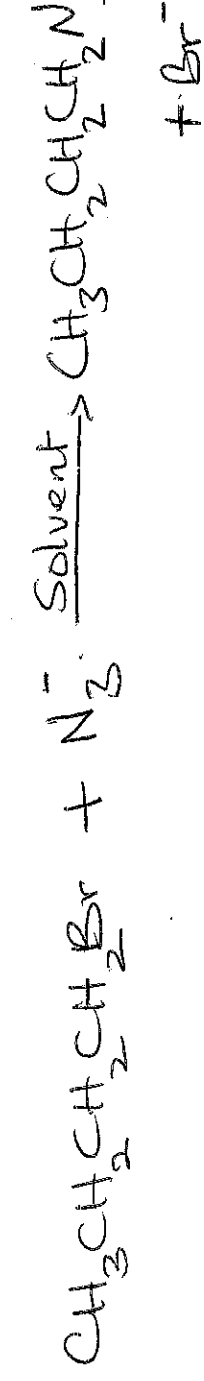
nucleophile and lowering its energy, ~~at~~ H^+ $\bar{\text{e}}$ form hydrogen bonding to $\bar{\text{e}}$ nucleophile, and thereby stabilizing it.



In contrast to protic solvents, aprotic solvents increase the rate of $\text{S}_{\text{N}}2$ reactions by raising the energy of the nucleophile. ~~For~~ eg of valuable aprotic solvents acetonitrile (CH_3CN), dimethylformamide (DMF), dimethyl sulfoxide ($(\text{CH}_3)_2\text{SO}$) and hexamethylphosphoramide, $[(\text{CH}_3)_6\text{P}=\text{O}]$ (HMPA).

This solvents can dissolve many salts b/c of their high polarity,

but they ~~are~~ ^{tend} to solvate metal cation rather than the nucleophilic anion. As a result the bare unsolvated anions have a greater nucleophilicity, and S_N2 rxns take place at a faster rate.

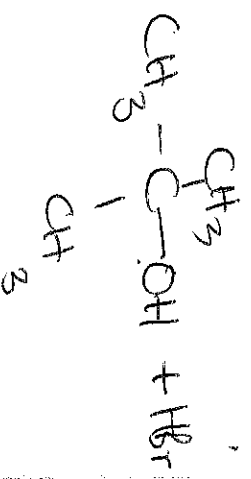
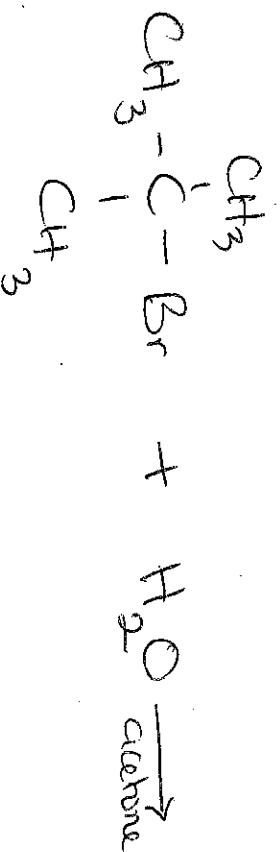


Solvent = HMPA	CH_3CN	DMF	DMSO	H_2O	CH_3OH
	5000	2800	1300	7	1
Relative reactivity	200,000				
More $\leftarrow$ reactive	Reactivity as Solvent				
	Less reactivity				

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The S_N1 Reaction

When proper conditions are chosen, tertiary haloalkanes will readily undergo $\text{S}_{\text{N}}1$ nucleophilic substitution reactions. eg



2-bromo-2-methyl
propane
(tert-butyl bromide)

2-methyl-2-
propanol
(tert-butyl alcohol)

Other nucleophilic rxns occur ~~at~~ tertiary carbon and non-basic nucleophiles.

These rxns occur ~~to~~ under diff conditions with treat-

of S_N2 reactions.

rate law = first order

rxn depends only on the

concentration of the haloalkene

(alkyl halide) and is independent of the conc of nucleophile

$$\text{Rate} = k [(\text{CH}_3)_3\text{Br}]$$

Second dlf from S_N2 -rxn is the stereochemistry of the 2

rxns.

S_N2 occurs by inversion of configuration at rxn site.

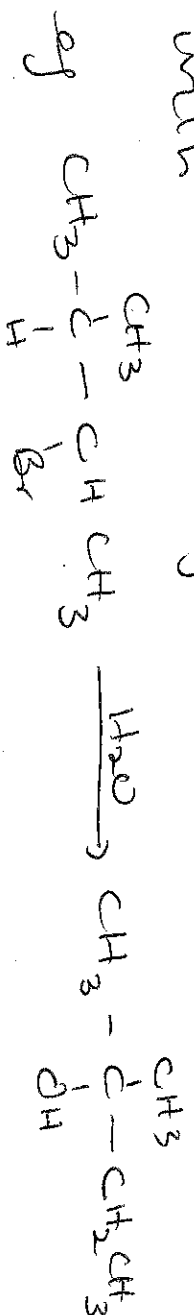
— S_N1 ~~occurs~~ optically achieve 3°; 2° alkyl derivatives ^{produce} a mixture of enantiomers; ~~occurs with retention of configuration~~

Reason: Carbocation is not ~~comp~~^{free} of its halide ion counterpart when it is attacked by the nucleophile. The anion of the leaving group shields one side of the carbocation, and the nucleophile captures the carbocation faster from the side opposite the leaving group. More inverted configuration is formed than products of retained configuration.

Carbocation rearrangement in S_N1 reactions

Products formed sometimes

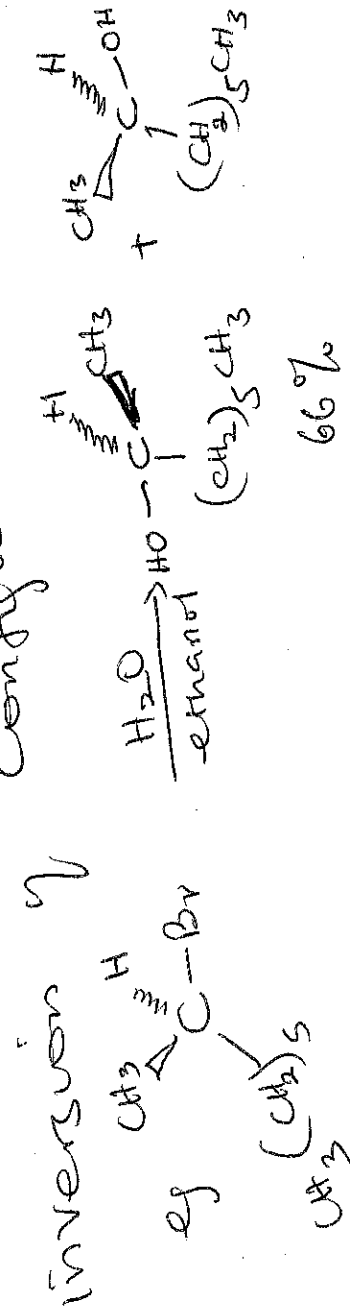
with rearrangement of Carbocation



(40)

The carbocation formed is achiral. Achiral b/c the 3 bonds to the charged carbon lie in the same plane. -CH is also the plane of symmetry ~~for~~ for carbocation.

A symmetrical carbocation should react with a nucleophile at the same rate at either of the faces, thus giving a racemic mixture ~~of~~ -CH is optically inactive though rarely observed in practice, b/c products are normally predominantly with inversion of configuration.



There is partial loss of optical activity.

(41)

Solvent effects

more apparent in S_N1 reactions

In S_N1 reactions neutral alkyl halide ionizes in the rate determining step forming a carbocation - halide ion pair.

A polar solvent will interact more with the polar transition state than with the starting alkyl halide.

water, alcohol, carboxylic acids are classified as protic solvents, b/c they all have

hydroxyl groups that make them capable of participating in hydrogen bonding, which may be to the leaving group or to the nucleophile.

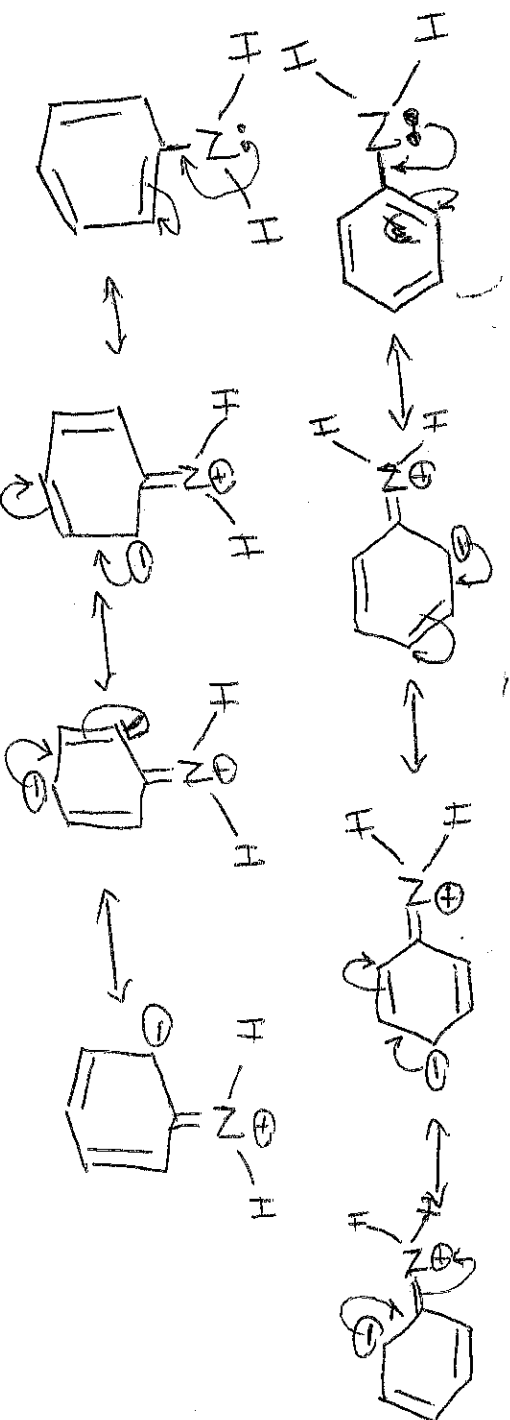
When the hydrogen bond is to the nucleophile it has

no effect on the rate of S_N1 anion.
however it reduces the rate
of substitution by the S_N2 mechanism
b/c by stabilizing the nucleophile
it lowers its energy and
makes it less reactive.

Polar aprotic solvents has
larger effects on the rate of
substitution by S_N2 mechanism.
Aprotic solvents lack $-OH$ groups
and cannot form hydrogen bonds
to the nucleophilic anion.
~~Non-solvated~~ nucleophiles
are more powerful than highly
solvated ones.

+M effects

if $\bar{e}$ es a lone pair of electrons on an atom directly attached to $\bar{e}$ benzene ring as, for example, in $\bar{e}$ case of aniline, mesomeric ~~off~~ interaction $\bar{e}$ ring takes place as follows

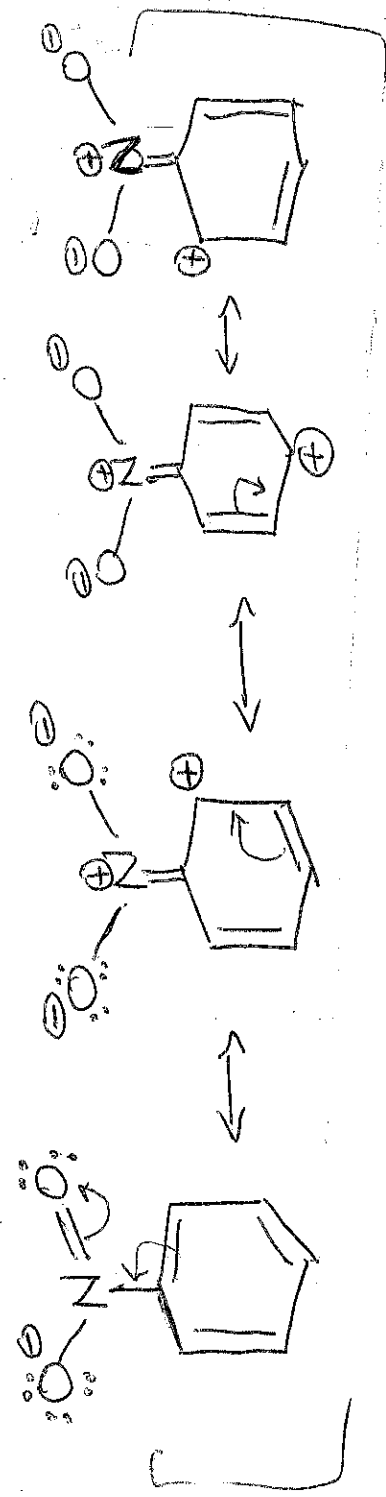


From $\bar{e}$ contributing structures it may be seen that $\bar{e}$ nitrogen atom partially gives $\wedge$ its lone pair of electrons to $\bar{e}$ benzene ring, th as a result (causes) requires an increased electron density (+M effect) Similarly contributing structures can be drawn for phenol or chlorobenzene. The greater $\bar{e}$ electronegativity of $\bar{e}$ atom attached to $\bar{e}$ benzene ring $\bar{e}$ more reluctant it is to bear a +ve charge

and in consequence the +M effect is smaller. The effect decreases in $\bar{e}$ order $\text{NH}_2 > \text{OH} > \text{Cl}$

-M effect

If substituent contains a double bond which is conjugated with benzene ring and it has a strongly electron-withdrawing atom at its end, further from benzene ring, then electrons are withdrawn from benzene ring, thus lowering electron density on benzene ring (-M effect). An example is provided by the nitro group.



Similar contributing structures can be drawn for in the case of, for example, alkyl phenyl ketones, benzenesulphonic acid and benzonitrile.